

SHOULD THIS CASE BE ADAPTED? PREDICTION FRAGMENTATION CONTROLS TEST-TIME ADAPTATION

Lili Wang, Jing Li, Xiaowen Sun, Xiangyu Hu, Zhuangzhuang Gu, Jian Liu, Srihari Nelakuditi, Yan Tong

ABSTRACT

Episodic test-time adaptation (TTA) resets a frozen segmenter to source weights M_0 on each case and adapts for a fixed number of steps. A fixed horizon conflates a cohort-level question, *how far to adapt*, with an irreducibly per-case one, *whether this case should be adapted at all*. Cohort means average over that per-case decision: on cross-vendor cardiac MRI the mean Δ Dice from adaptation is statistically indistinguishable from zero while 58.7% of cases are individually made worse. We quantify this harm as *harmful accepted area* (HA), the harmful fraction of the edited area a controller actually deploys. Held-out tuning gives a stronger baseline than a fixed horizon, but the budget it selects does not transfer on either of the two main medical benchmarks, and no global budget can condition on the case. We show that prediction fragmentation—the disagreement geometry between M_0 and the adapted mask M_k —predicts HA with no labels and no additional backward passes at decision time, at comparable strength on three benchmarks (Spearman ρ 0.50–0.60), at a quarter of gradient-norm’s latency. A case-level router built on it cuts HA from 0.228 to 0.139 on a benchmark that took no part in its design, with that design frozen and only its cut-points recalibrated there. On the cardiac benchmark the design was selected on, the same router cuts HA from 0.129 to 0.013 at matched Dice and 1.10 deployed updates, against the retrospective-best budget identified post hoc from evaluation labels, and reduces that 58.7% to 20.0%—an upper bound we quantify. Where the retained cases are not net-helped (as on prostate), the router still cuts HA but concedes accuracy, a boundary we report explicitly. Thresholds are calibrated once on a labeled split disjoint from evaluation; deployment decisions use no labels and no gradients. The template ports across architecture and domain (nnU-Net→SegFormer, Cityscapes→ACDC) with coordinate, thresholds and per-bucket actions instantiated per domain.

1 INTRODUCTION

Test-time adaptation (TTA) updates a trained model on the test data itself, with no source data and no labels, to recover accuracy lost to distribution shift. Episodic TTA is its per-case form and is increasingly deployed on frozen segmentation models: reset to source weights M_0 , adapt for a short horizon on each new case, and return the adapted state M_k (Wang et al., 2021; Niu et al., 2022; Wang et al., 2022). In practice that horizon is fixed in advance and applied uniformly to every case, which conflates two decisions. *How far to adapt* is a cohort-level question, answerable by tuning the budget on held-out data. *Whether this case should be adapted at all* is irreducibly per-case, and no global budget can make that decision. That per-case axis is where the reliability of episodic TTA is settled, and cohort means do not show it.

Mean overlap hides harm. Adaptation can raise average Dice / mIoU while, in localized regions, editing the prediction from correct to incorrect. We call the fraction of edited area that becomes locally worse the *harmful accepted area* (HA). For instance, on cross-vendor cardiac MRI cohort-mean Δ Dice is statistically indistinguishable from zero while 58.7% of cases are individually made worse (Fig. 1); HA tracks a clinically used boundary metric we never optimize, and is non-monotone in the budget (Sec. 3). A held-out-tuned budget is a stronger baseline than the fixed horizon, and we adopt it. But the reliability-optimal budget does not transfer from calibration to evaluation on either of the two main medical benchmarks, while thresholds fit there are applied unchanged (Sec. 3.1).

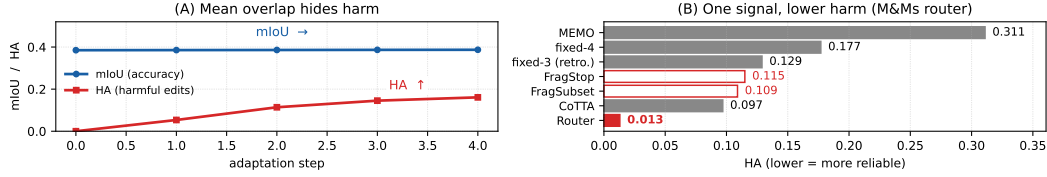


Figure 1: **One signal, from problem to payoff.** (A) mIoU flat while HA rises (ACDC rain, $1r\ 10^{-4}$, $K=4$; the aggressive budget of Sec. 5.3 is steeper still). (B) M&Ms router attains the lowest HA at matched Dice (ours in red, solid), a per-deployment result, not universal dominance (Sec. 5.2). Against the retrospective fixed-3 only the router separates cleanly; the two outlined budget-axis controllers are marginal or not separated (Sec. S2.8). Cohorts, budgets and panel (A)’s pooling convention: Secs. S1.9, S1.7.

What a global budget cannot do at all is condition on the case. We therefore read the disagreement geometry between M_0 and M_k as a per-case signal, *prediction fragmentation*. It uses predictions only, with no labels and no gradients at decision time. Among the read-only signals we evaluate, it is the strongest predictor of HA by pooled correlation, at a quarter the latency of gradient-norm and negligible extra memory. It also drops into an episodic TTA routine without changing the adaptation objective. Our main instance is a case-level router: after one step it sends each case to rollback, a curtailed adaptation, or the least restricted one. The same signal also instantiates a stop rule and a modulation of *which* parameters adapt, reported as ablations.

Contributions. We define HA and show that mean overlap hides the reliability failure of fixed-budget episodic TTA and that the HA-optimal budget does not survive a change of split on M&Ms or prostate OOD-all (Sec. 3). Prediction fragmentation predicts that harm without labels or gradients at decision time, at comparable strength on three benchmarks (Sec. 4.1). A case-level router built on it cuts the individually harmed fraction at matched Dice on cardiac and ports across architecture and domain, with coordinate, thresholds and per-bucket actions instantiated per domain. Frozen but for its cut-points, the design transfers to a benchmark it was not selected on (Secs. 4.2, 5.1–5.3).

2 RELATED WORK

Test-time adaptation and fail-safe mechanisms. Episodic TTA optimizes, without source data or labels, proxy objectives such as entropy minimization (Wang et al., 2021), batch-norm statistics alignment (Schneider et al., 2020), augmentation-consistent teacher–student updates (Wang et al., 2022), marginal-entropy minimization over augmentations (Zhang et al., 2022), or a self-supervised auxiliary task (Sun et al., 2020); EATA (Niu et al., 2022) adds sample selection and anti-forgetting, and SAR (Niu et al., 2023) filters updates by entropy and sharpness. These methods change *what* objective adapts M_0 , and their sample selection decides *which samples* to adapt on. We take their trajectories as given and ask a downstream question they do not address: given a materialized trajectory, should *this* case keep its adaptation? A parallel line monitors adaptation rather than modifying it. AETTA (Lee et al., 2024b) estimates accuracy from dropout-inference disagreement to trigger model resets, building on agreement-based OOD accuracy estimation (Baek et al., 2022); TEGDA (Zhou et al., 2025) proposes a confidence-calibrated dropout-agreement metric for the same lack of ground-truth monitoring; and Schirmer et al. (2025) raise an alarm when an adapting model’s risk drifts over a continuous test stream. The last is closest in framing and differs in unit and output: under per-case reset there is no drift to accumulate, and a monitor emits an alarm where we emit an action on this case’s prediction. We differ from all three in the signal, which needs no stochastic forward passes. FOA (Niu et al., 2024) removes backpropagation from the update itself, an axis orthogonal to our backprop-free *control* signal and one that matters on its own, since rankings shift under a compute budget (Alfarra et al., 2024).

Label-free quality assessment and disagreement signals. Estimating segmentation quality without annotations is a long-standing problem in medical imaging (Audelan & Delingette, 2021). Under domain shift, existing signals derive from per-pixel confidence, stochastic-pass disagreement (Lee et al., 2024b; Zhou et al., 2025), or model-space statistics: GraTa (Chen et al., 2025) corrects gradient

directions via alignment between pseudo- and auxiliary gradients, and DeYO (Lee et al., 2024a) augments entropy with a shape-sensitivity score. Model-space controllers change *how* an update is formed, so the two axes compose rather than compete. A fragmentation router wraps a GraTa-optimized trajectory as it wraps ours, though there a matched-quota random control reproduces most of what it buys (Sec. S12). We include GraTa’s alignment as a signal-level baseline (Sec. 4.1). Our signal reads the connected-component geometry of the disagreement between M_0 and M_k , a per-case, backprop-free statistic. Early-exit methods (Teerapittayanon et al., 2016; Graves, 2016) halt inference; our stop rule halts *adaptation* of a frozen predictor.

Evaluation granularity and harm. Region-sensitive metrics such as Surface Dice (Nikolov et al., 2021), Boundary IoU (Cheng et al., 2021) and the Metrics Reloaded recommendations (Maier-Hein et al., 2024) address localized errors that aggregate overlap metrics can underemphasize, but all compare against ground truth offline. Harmful accepted area (HA, Sec. 3) shares that concern with a different reference: it scores the edits a controller actually deploys against the source prediction M_0 .

3 PRELIMINARIES: THE RELIABILITY FAILURE OF FIXED-BUDGET TTA

Episodic TTA on a frozen M_0 produces predictions $M_1, \dots, M_T$. Let a, b, y be the per-voxel argmax labels of M_0, M_k and GT. On the medical benchmarks all three are first binarized to foreground, so only foreground/background edits enter D . On driving, the 19-class labels are used as is, so any label change does (Sec. S1.5). The disagreement mask is $D = \{x : a(x) \neq b(x)\}$; we extract connected components $\{R_j \subseteq D\}$ with $|R_j| \geq \tau_{\text{area}}$. A region is *harmful* if $\text{err}(b, R_j) > \text{err}(a, R_j)$, where $\text{err}(m, R) = \frac{1}{|R|} \sum_{x \in R} \mathbb{I}[m(x) \neq y(x)]$. Deployment HA is the harmful fraction of edited area:

$$\text{HA}(M_k) = \sum_{j: \text{harmful}(R_j)} \frac{|R_j|}{|D|}, \quad \text{HA}(M_k) = 0 \text{ when } D = \emptyset. \quad (1)$$

Two conventions make HA a deployment metric. It is *M_0 -referenced*: every controller’s output is scored against the same source $a=M_0$, so “harmful edit” means the same thing for every method. The edited set D , by contrast, is the one that controller actually deploys (M_0 vs. M_4 for fixed-4, vs. M_T for a stop, the post-policy edits for a router). HA is therefore comparable across methods, but its denominator $|D|$ moves with the gate, a property with consequences in both the budget (Sec. 3.1) and the gate (Sec. 4.2). And cohort HA is the unweighted mean of per-case HA, so a small edit set counts as much as a large one.

HA=0 means “not judged harmful”, not “judged harmless”. Two things score zero. Reverting to M_0 deploys no edits ($D=\emptyset$), so full rollback attains HA=0 trivially and HA is not meaningful alone. It is therefore always reported beside Dice/mIoU and deployed steps, with edit coverage, matched-rollback controls and the composition of the declined half in Secs. S6, S3, S5. A case whose disagreement set is non-empty but has no component of at least τ_{area} (16 voxels on medical, 200 px on ACDC) also scores zero; on cardiac this covers 376 of 920 (case, step) rows. The absolute value depends on that threshold; on cardiac, the router’s rank and the sign of its margin do not, across an eightfold change in it (Sec. S8).

HA is a reliability lens, not a loss we optimize or a replacement for Dice/mIoU (Sec. 5.4). It tracks a boundary metric we never target: Spearman $\rho=0.62$ with HD95 of M_k on prostate OOD-hard ($n=34$) while cohort-mean ΔHD95 is -13 mm. The failure is visible case by case on a clinically used metric even as that metric’s own mean improves (Sec. S1.10).

3.1 HA IS NOT MONOTONE IN THE ADAPTATION BUDGET

Because HA is a *fraction* of a controller-dependent edit set, it need not increase with the budget: additional steps enlarge the denominator $|D|$ as well as the harmful numerator. Table 1 sweeps the fixed budget on the same trajectories used throughout. On all three medical splits HA is minimized at $K=3$ and is higher at $K=4$, so the default fixed horizon is not the strongest fixed baseline on these trajectories. Dice, meanwhile, moves little across the sweep (0.002, 0.007 and 0.015 on the three rows). We therefore adopt the held-out-tuned budget, not the fixed horizon, as our primary point of comparison. (On ACDC rain under TENT the ladder rises monotonically over $K \leq 8$; the shape belongs to EATA on this horizon, not to HA.)

Split	$K=1$	$K=2$	$K=3$	$K=4$
M&Ms ($n=230$)	0.186	0.164	0.129	0.177
Prostate OOD-all ($n=93$)	0.242	0.211	0.171	0.262
Prostate OOD-hard ($n=34$)	0.290	0.309	0.257	0.318

Table 1: **HA vs. fixed budget** (EATA trajectories, anchor variant in Sec. S1.3; Dice omitted, it varies by 0.002, 0.007 and 0.015 across the three rows). Per-split holdout→eval selection is in Sec. S1.4.

Benchmark (objective, budget)	Signal	ρ vs HA	$\rho(\Delta\text{signal}, \Delta\text{HA})$	backprop
Prostate MRI, $n=124$ (EATA, lr 5×10^{-4} , $K=4$)	n_{reg}	0.594	0.464	No
	δ	0.557	0.515	No
	entropy	0.119	-0.290	No
	MSP	-0.161	0.278	No
	grad-norm	0.270	-0.035	Yes
	GraTa-cos	-0.093	-0.009	Yes ($\times 2$)
Cardiac MRI, $n=230$ (EATA, lr 5×10^{-4} , $K=4$)	n_{reg}	0.597	0.455	No
	δ	0.516	0.347	No
	entropy [†]	0.032	-0.006	No
	MSP [†]	-0.012	0.017	No
	1-agreement [†]	0.264	0.051	No ($8 \times$ fwd)
	grad-norm	0.146	0.035	Yes
Driving, $n=406$ (TENT, lr 10^{-4} , $K=4$)	n_{reg}	0.503	0.290	No
	δ	0.478	0.238	No
Driving, $n=306$ (TENT, lr 2×10^{-3} , $K=8$)	n_{reg}	-0.147	0.046	No
	δ	0.139	0.187	No

Table 2: Harm-predictive comparison across three benchmarks (EATA rows use the anchor variant of Sec. S1.3). Correlations pool (case, step) pairs with step ≥ 1 ; Δ differences consecutive steps within a case. Driving is ACDC pooled over conditions, reported at both budgets because the association inverts at the aggressive one. [†]From a verification rerun of the same cardiac trajectories, whose shared-signal correlations match the archived run to ± 0.002 (Sec. S9); the archived run stays authoritative for every main-table number. The GraTa-cos comparison is available on prostate only. MSP is maximum softmax probability. Clustered intervals, the per-step decomposition and the case-level ranking AUROCs are in Sec. S4.

But the optimal budget cannot be selected in advance. Holdout selection gives $K=1$ on prostate and $K=2$ on M&Ms against an evaluation optimum of $K=3$ on both, and under joint bootstrap ($B=5000$) the two optima agree on only 7.1% and 0.3% of resamples. Every selection rule we tried makes the same choice, while the thresholds and routing cut-points fit on the same splits are applied unchanged (Secs. S1.1, S1.2, S1.4).

A global budget cannot decide case by case. On M&Ms the cohort-mean ΔDice from fixed-budget adaptation is $+0.00006$ (95% CI $[-0.0012, +0.0014]$, i.e. $+0.000$ at the precision of every table here), yet 58.7% of cases are individually made *worse*. Tuning K moves the mean but cannot separate these two populations, because it does not condition on the case. That axis is what the rest of the paper addresses.

4 METHOD

4.1 PREDICTION FRAGMENTATION AS A HARM-PREDICTIVE SIGNAL

At step k , define $s_k = \phi(M_0, M_k) = (n_k, \Delta n_k, \delta_k, \Delta \delta_k, \bar{H}_k)$, where n_k is the number of disagreement regions, δ_k the disagreement ratio, $\Delta n_k = n_k - n_{k-1}$, and $\bar{H}_k$ the mean predictive entropy on disagreement voxels. We use “fragmentation” as shorthand for this family of disagreement-geometry statistics; n_{reg} and δ are read from the same two masks (Table 2), and which one is deployed is a per-domain choice (Sec. 4.2, Sec. S10). On a per-site holdout we fit z-score statistics $(\mu_n, \sigma_n), (\mu_\delta, \sigma_\delta)$

and define a scalar fragmentation risk

$$r_k = \frac{1}{2}z(n_k; \mu_n, \sigma_n) + \frac{1}{2}z(\delta_k; \mu_\delta, \sigma_\delta). \quad (2)$$

What the signal is, and what it is not. Fitting (μ, σ, τ) is done *once* on a held-out labeled split disjoint from the online test stream, exactly as TENT/EATA/CoTTA hyperparameters are. At deployment no labels and no gradients are used, so the label-free/backprop-free claim refers strictly to per-case decision time. The signal operates at the *case* level and not below it: a higher n_{reg} reflects many mutually unrelated changes rather than one coherent revision, and it is that aggregate instability that correlates with harm. We do *not* claim an individual harmful edit is geometrically identifiable; a region-level analysis reverses the “harmful edits are small debris” picture on prostate (Sec. S1.11).

The association, and how it compares. Step 0 is excluded because HA=0 there by construction, and every value comes from stored per-step statistics (exceptions in Sec. S9). Fragmentation is largely complementary to confidence: on prostate its level correlation with entropy and MSP is near zero (0.03 and -0.10 ; Sec. S1.6), as expected if harm is carried by the distribution of disagreement rather than by per-pixel certainty, since a wrong edit can be made confidently. Against HA it ranks highest of the read-only signals on both medical benchmarks by pooled correlation. The association also replicates at comparable budgets: n_{reg} reaches $0.594 / 0.597 / 0.503$ on prostate, cardiac and driving, and the step-differenced column $0.464 / 0.455 / 0.290$, across nnU-Net and SegFormer, MRI and natural images, EATA and TENT. The comparison that motivates the signal survives the change of benchmark in our favor. Grad-norm falls from 0.270 to 0.146 between the two medical benchmarks while fragmentation holds near 0.59 , so the read-only signal separates *further* from the backprop one on the second benchmark (cardiac). GraTa’s alignment (Chen et al., 2025) is near-orthogonal to harm despite two backward passes per step; we compare at the signal level since alignment governs *how* an update is formed, not *how many*. Ranking cases is only possible for the harm HA measures. No signal we test predicts case-level Dice degradation (AUROC 0.37 – 0.56 ; grad-norm on cardiac is inverted, $[0.30, 0.45]$), while every signal ranks HA-level harm (0.62 – 0.72 ; MSP inverted; Sec. S4). On Prostate158 neither δ nor n_{reg} does ($0.475, 0.506$; Sec. S3).

4.2 COMPILING THE SIGNAL INTO CONTROLLERS

The signal is read through a different reduction in each controller: FragStop halts on a rise in n_k with nothing calibrated, the ACDC τ -stop thresholds the single-term $z(n_k)$, FragSubset thresholds the two-term risk r_k of Eq. 2 at zero, and the router splits one per-domain coordinate $u_k = g_d(s_k)$ at holdout tertiles (Sec. S1.7). Eq. 2 therefore enters at one point only, the parameter-subset modulation. Sec. 3.1 showed the budget axis cannot be settled in advance, so we take the *per-case* action as primary and report the budget-axis actions as ablations that establish the signal compiles.

COQR Router. Case-level online quantile routing (COQR) acts on the axis no global budget reaches: whether *this* case should keep its adaptation. The skeleton is one rule: score the case after step 1, split the score at holdout-calibrated tertiles (which need not divide the evaluation stream evenly; Sec. S1.7), and attach an action of decreasing severity to each bucket, from rollback through a curtailed adaptation to the least restricted one (Table S1). Rollback returns M_0 and deploys no edits. What the other two buckets do is instantiated per domain. On the two medical benchmarks each bucket deploys a fixed depth: 3 and 2-then-shrink ($\alpha=0.5$) on M&Ms, 1 and 1 on prostate. The partition is three-way; prostate’s action set is two-way. On driving, both buckets run the stop rule below within a per-bucket cap, and their depth varies by case. The fragmentation state therefore enters the bucket assignment in every domain, and the depth only on driving. The scoring coordinate is likewise per domain (δ on M&Ms and ACDC, n_{reg} on prostate). We do not claim one coordinate serves every setting: Sec. S11 gives a backbone on which δ fails and n_{reg} does not (on few scorable cases). Sec. S10 documents how each was chosen, including four choices made with an evaluation split visible. COQR is *not* a learned meta-controller, and no evaluation label enters the per-case decision: the cut-points are fixed once on the held-out split, then applied unchanged online. The decision needs no gradient, since it reads s_k from the two masks. But it is taken *after* one adaptation step, so a rolled-back case still pays that step. What rollback saves is the remaining budget, and the *Steps* column counts deployed updates (Sec. S1.7).

Two budget-axis instantiations. **FragStop** halts at the first rise $\Delta n_k > 0$, evaluated from $k \geq 2$, so at least one step is always kept and the rule is the most conservative feasible instance thereafter.

Method	Dice	HA	Steps
M_0 (no adapt, ref.)	0.849	0.000	—
fixed-2 (<i>deployable</i>)	0.850	0.164	2.00
fixed-3 (<i>retrospective</i>)	0.850	0.129	3.00
fixed-4 (<i>default</i>)	0.849	0.177	4.00
FragStop (ours)	0.850	0.115	3.43
FragSubset (ours)	0.850	0.109	3.43
COQR Router (ours)	0.849	0.013	1.10
MEMO	0.841	0.311	4.00
CoTTA	0.849	0.097	4.00

Table 3: M&Ms cardiac Vendor-B OOD ($n=230$, online); $k=1$ is in Table 1. Fixed- k rows are successive checkpoints of one episodic EATA trajectory. The router rolls back 130 of 230 cases; *Steps* is the deployed depth (executed updates; Sec. S5). The FragSubset modulation never fires on this benchmark; its difference from FragStop traces to four cases (Secs. S1.3, S2.3).

Method	Dice	HA	Steps
M_0 (no adapt, ref.)	0.738	0.000	—
fixed-1 (<i>deployable</i>)	0.752	0.242	1.00
fixed-2	0.753	0.211	2.00
fixed-3 (<i>retrospective</i>)	0.751	0.171	3.00
fixed-4 (<i>default</i>)	0.746	0.262	4.00
FragStop (ours)	0.749	0.183	2.77
FragSubset (ours)	0.755	0.166	2.68
COQR Router (ours)	0.739	0.099	0.55
MEMO	0.740	0.278	4.00
CoTTA	0.732	0.122	4.00

Table 4: Prostate MRI OOD-all ($n=93$), online; conventions as in Table 3. Every method is run on the identical case set. The router rolls back 42 of 93 cases. The harder $n=34$ subset and the ablation configuration names are in Secs. S2.1 and S2.3.

FragSubset adds modulation on top of that stop: it updates decoder normalization only when $r_k > 0$ and all normalization otherwise, so high fragmentation narrows *what may move* rather than reweighting the objective. Whether that condition fires at all is a property of the benchmark’s risk distribution rather than of the rule: it fires on 14–24% of prostate steps and on no M&Ms step (Sec. S2.3). The two other members of the modulation family, and the configuration names, are in Sec. S2.3. Both improve on the *deployable* fixed budget; against the retrospective one only FragSubset on M&Ms separates, and marginally (Sec. 5.4); we report them to establish that one statistic instantiates stopping, modulation and routing, not as gains over a tuned budget.

HA cannot be certified directly. HA’s denominator moves with the gate, which defeats monotone conformal certification: the risk must be monotone in the threshold and deployment HA is not. Fixing the denominator gives a surrogate that admits the standard guarantee unchanged. Sec. S2.5 reports the construction and the gap between that surrogate and deployed HA, $1.7\times$ to $4.0\times$ across all seven feasible operating points.

5 EXPERIMENTS

All methods share one episodic loop and one calibration rule. Every threshold is fit once on a labeled holdout disjoint from the evaluation stream: (μ, σ, τ) , the routing cut-points and $\hat{\lambda}$. Ground truth enters only offline (Sec. S1.9 lists where). Our primary comparison is against the *deployable* budget, the fixed k that holdout selects.¹

5.1 CARDIAC MRI: MATCHED DICE, LOWER HARM

On the cross-vendor cardiac split of M&Ms (Campello et al., 2021) with a frozen nnU-Net (Isensee et al., 2021) (deployable budget $k=2$, retrospective $k=3$; Sec. 3.1), the router attains the lowest HA of any adaptive method (0.013) at 1.10 deployed updates per case. It clears both references: Tier-A against fixed-2, and Tier-B against the retrospective fixed-3 ($\Delta\text{HA} -0.116 [-0.152, -0.081]$), a budget that could only have been identified post hoc, with evaluation labels. The router rolls back 130 of 230 cases, and of the 100 it retains only 17 deploy an edit with a component above the size threshold. Retained-only HA is 0.031, and the deployed edit set covers $\delta=3\times 10^{-5}$ over all 230 cases, an order of magnitude below the fixed budgets (3.2×10^{-4} at fixed-3). The headline is the abstention

¹Tiers: **A** vs. the deployable budget ($k=2$ on M&Ms, $k=1$ on prostate), **A'** vs. fixed-4, **B** vs. the retrospective eval-best k (not deployable), **C** descriptive. Intervals are $B=5000$ paired resamples; P denotes $P(\Delta\text{HA}<0)$ over them; full protocol in Sec. S1.9.

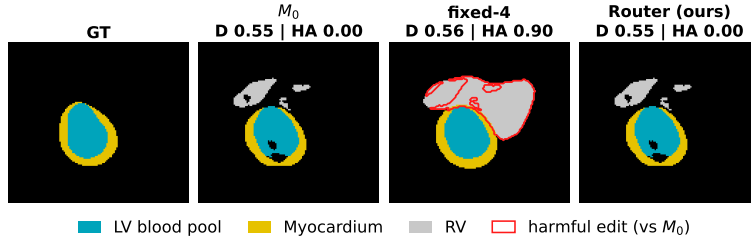


Figure 2: Harm invisible to Dice (M&Ms, router, A8E1F4_frame09). L→R: GT (evaluation reference only); M_0 ; fixed-4, with every harmful accepted edit outlined in red; ours, which routes this case to rollback and so returns M_0 unchanged. This slice has no RV in GT, so the foreground Dice we report is blind to the false-RV expansion HA flags. Panel metrics are slice-level; selection protocol in Sec. S1.15.

Split	Method	helped ($\mu_{\Delta+}$)	hurt ($\mu_{\Delta-}$)	rolled back	overall μ_{Δ}
M&Ms	fixed-4	41.3 (+0.007)	58.7 (−0.005)	—	+0.000
M&Ms	Router	23.5 (+0.001)	20.0 (−0.001)	56.5	+0.000
Prostate	fixed-4	51.6 (+0.046)	48.4 (−0.031)	—	+0.009
Prostate	Router	21.5 (+0.019)	29.0 (−0.011)	45.2	+0.001
ACDC night	TENT ₈	42.5 (+0.028)	57.5 (−0.042)	—	−0.012
ACDC night	Router	34.0 (+0.018)	25.5 (−0.030)	40.6	−0.002

Table 5: Case-level decomposition of Δ against M_0 (% of all cases; μ_{Δ} at three decimals, which is what hides the 58.7%). On M&Ms and night the router retains more helped than hurt; on prostate it does not (bold marks the larger of the two on each router row). Remaining ACDC conditions and the prostate zero- Δ convention are in Sec. S1.12.

component Sec. 3 says to look for: a small, well-chosen edit set, not a large harmless one (Sec. S6). It is not bought with overlap: Δ Dice vs. fixed-3 is -0.0008 $[-0.0022, +0.0006]$ and vs. M_0 at full precision $+0.000016$ $[-0.00007, +0.00010]$, the retained buckets nearly cancelling. On a stream with no net accuracy to gain, that is the claim: the router recovers the source model’s accuracy *and* its safety without being told in advance which cases adaptation would have hurt. The objection that M_0 already attains that pair here is the per-case form of Sec. 3.1’s budget objection. *Not adapting* is the right action on this stream, but which streams it is right on is known only after the fact. The sign is not even constant within a dataset: at one learning rate and one backbone the router sits above M_0 on ACDC fog, rain and snow ($+0.007$ to $+0.010$ mIoU) and below it on night (-0.0015 ; Sec. S5). No cohort-level rule reads that off a stream in advance. Fig. 2 shows fixed-4 buying $+0.01$ slice Dice at HA 0.90 while the router returns M_0 . Remaining pairwise intervals are in Sec. S2.8.

The same design, frozen, on a benchmark it never saw. Because the M&Ms design was selected with evaluation-split visibility (Sec. S10), we test it where that exposure cannot act: coordinate, tertile structure and per-bucket actions frozen, cut-points refit on a 32-case holdout, on Prostate158 (Adams et al., 2022) ($n=126$), which took no part in the sweep. The frozen design cuts HA against that benchmark’s own retrospective-best budget from 0.228 to 0.139 (-0.089 $[-0.124, -0.055]$) at a Δ Dice interval containing zero, and sits above M_0 . Its relative HA reduction there (39%) matches prostate OOD-all (42%) rather than M&Ms (90%); Sec. S10 gives the panel and three accounts of that gap; at matched quota and depths the ranking there is not better than random (Sec. S3).

5.2 PROSTATE MRI: WHERE THE PER-CASE TRADE IS UNFAVORABLE

On the multi-site prostate benchmark (Liu et al., 2020), again with a frozen nnU-Net (Table 4), the router still cuts HA but fails our own Dice-matched standard, and we report that as the boundary. It clears both references (Tier-A vs. fixed-1, Δ HA -0.142 $[-0.194, -0.094]$; Tier-B vs. fixed-3, -0.072 $[-0.139, -0.007]$) at 0.55 deployed updates (1.00 SGD steps counting the discarded probe, a quarter of the default budget). The cost is on the other axis: Δ Dice vs. fixed-1 is -0.0130

$[-0.0252, -0.0022]$, an interval that excludes zero, unlike every Dice comparison on M&Ms; at 0.739 the router sits at the un-adapted source, having rolled back 42 of 93 cases.

Table 5 says why. The router separates two opposed populations rather than shifting an average: on M&Ms the individually harmed fraction falls $58.7\% \rightarrow 20.0\%$ at the same cohort mean. The router also makes the boundary a condition rather than a split we happen to lose on. On both benchmarks the signal orders source-model difficulty: rolled-back cases have a weaker M_0 than kept ones (0.836 vs. 0.866 on M&Ms, 0.578 vs. 0.869 on prostate). What differs is the kept half: on M&Ms helped outnumber hurt and Δ Dice against the deployable budget contains zero ($-0.0015 [-0.0032, +0.0001]$); on prostate hurt outnumber helped and it does not. Ordering difficulty is not separating benefit from harm, and only the second buys reliability for free. On prostate fixed-4 is close to even over the whole split (48.4% hurt), so ranking by risk selects against benefit. The failure is testable in advance: the helped/hurt balance of the retained bucket can be read off a labeled holdout before deployment. Matched-bucket controls agree (Sec. S3): fragmentation beats random at the same rollback rate ($P=0.010$) but is indistinguishable from confidence-scored routing here, whereas on M&Ms only fragmentation separates from every alternative (Sec. 5.4). On the hardest $n=34$ subset MEMO attains the best HA and, among the non-fixed methods, the best Dice, at full budget with extra backprop, and the router is not reportable (the low bucket is empty; Sec. S2.1).

5.3 TRANSFER AND THE SIGNAL’S BOUNDARY: CITYSCAPES $\rightarrow$ ACDC

We transfer the controller template to driving segmentation, changing backbone, domain and objective: a frozen Cityscapes-pretrained SegFormer-b0 (Xie et al., 2021; Cordts et al., 2016) (19 classes), the ACDC adverse conditions (Sakaridis et al., 2021) as targets, episodic TENT-style adaptation, the same class-agnostic HA and signal, thresholds and tertiles calibrated on ACDC *train* and evaluated on *val*. Under a budget large enough to over-adapt ($\text{lr } 2 \times 10^{-3}$, $K=8$) the ladder on rain is monotone rather than U-shaped: over $k \in \{1, 2, 4, 8\}$ mIoU moves $0.378 \rightarrow 0.385 \rightarrow 0.389 \rightarrow 0.387$ while HA rises $0.264 \rightarrow 0.327 \rightarrow 0.385 \rightarrow 0.448$ (all steps in Sec. S1.4). The HA-optimal budget is simply $K=1$, and *train* $\rightarrow$ *val* selection recovers it in every resample. That poses the objection in its cleanest form: if the budget is both optimal and selectable, why route? Against fixed-1 the router still cuts HA by $-0.048 [-0.088, -0.012]$ at Tier-A, while the stop rule is significantly *worse* ($+0.047 [+0.025, +0.072]$), since it is defined against an over-adaptation that $k=1$ never produces.

Read against TENT₈, adding the stop rule cuts HA on every condition and routing cuts it again ($0.501 \rightarrow 0.375 \rightarrow 0.232$ on night; the other three in Sec. S2.7), at about one deployed update per case except on night (1.67), against $K=8$. The hard tertile again lands where M_0 is weakest. Night is the boundary: M_0 is near-collapsed, TENT₈ degrades *below* it (0.182 vs. 0.194), and the router’s 0.193 exceeds TENT₈ only because TENT falls under the source (damage control, not gain). The same regime is why CoTTA’s low HA is not a stronger safety result: its mIoU sits at M_0 on all four conditions, as its Dice does on M&Ms and below M_0 on prostate. Per-condition thresholds assume the condition is known at deployment; we did not run the pooled alternative (Sec. S1.7).

Where the signal stops working. At this budget the n_{reg} association with HA inverts to $-0.147 [-0.238, -0.053]$ while δ stays positive (0.139; Δ 0.187). Restricting the same trajectories to $k \leq 4$ does not restore n_{reg} (-0.054 rain, -0.222 snow, -0.142 night), so the cause is update size, not trajectory length. A near-collapsed M_0 makes disagreement large-scale re-guessing rather than sparse local edits, and counting components stops being informative once almost everything is recounted. This rules n_{reg} out as the driving coordinate. The statistic itself is computed without labels, but its correlation with HA, and hence this boundary, is not. The boundary is consistent with a matched control on δ : at matched rollback quota, per-case selection here is not better than random overall, fog excepted (Sec. S3), so the routing increment at this budget comes chiefly from the quota itself.

5.4 ABLATIONS, CONTROLS AND COST

Stopping and modulation. Both budget-axis controllers lower HA against the deployable budget (Tables 3, 4). Paired bootstrap confirms it for FragSubset ($-0.076 [-0.113, -0.041]$ vs. fixed-1 on prostate, $-0.055 [-0.084, -0.028]$ vs. fixed-2 on M&Ms, where the modulation does not fire and the reduction is the stop rule’s). Against the retrospective optimum the separation largely disappears (FragSubset vs. fixed-3 on OOD-all: $-0.006 [-0.046, +0.035]$; the rest in Sec. S2.8). Modulation

beyond stopping is not established by these experiments (OOD-hard: Wilcoxon $p=0.016$, Holm-adjusted 0.065, CI crossing zero). On OOD-hard, against fixed-4 the variants are ordered in point estimate (Sec. S2.3): parameter subset -0.042 , stop alone -0.033 , entropy weight -0.005 , objective switch $+0.024$. SAR (Niu et al., 2023) bounds the stop rule instead: it falls below the source from step one, leaving no early segment worth keeping (Sec. S2.4). The modulation rests on a narrow base: it fires on 14.1% of OOD-hard steps, never on M&Ms.

Controls and cost. At its own rollback rate on M&Ms, the router beats every equally conservative rule on HA (random, confidence- or agreement-scored routing) while staying within 0.001 on Dice (Sec. S3). The controllers never read HA; they act on fragmentation, which correlates with it at $\rho=0.516$ for the coordinate deployed here, not 1.0. Decision-time costs per signal are in Table S19.

6 DISCUSSION AND LIMITATIONS

What the controller presumes, and how to check it beforehand. Three conditions matter, two for the router and one for the budget-axis controllers, and each fails somewhere in this paper; only the first can be checked without labels. (i) *A usable M_0 whose edits are contiguous.* A near-collapsed M_0 turns the controllers into damage control (ACDC night). A transformer backbone edits in scattered voxels rather than regions, so δ stops carrying the signal while n_{reg} still does (on few scorable cases, Sec. S11). Under large modality shift the disagreement scale itself moves, by roughly $40\times$ on fundus (Sec. S1.14). These show in the geometry of the edits (Sec. S11). (ii) *A trajectory whose HA dips before it rises.* For the budget-axis controllers, per-case control is worth having because HA is non-monotone in k , and that shape depends on the objective and the horizon. It holds for EATA on the medical benchmarks and not on ACDC rain, where objective, backbone and budget all differ (Sec. 3.1); under two other objectives it reproduces in one of six cells, and there only formally; in the other five HA rises monotonically from $k=1$ (Sec. S12). A fixed- k ladder on the calibration split indicates which case one is in, though on prostate the two splits disagree (Sec. S1.4) and its HA needs labels. (iii) *A retained bucket that is net-helped.* Where it is not, the router still cuts HA but concedes accuracy (prostate, Sec. 5.2). This too can be read off the labeled calibration split.

What the headline number estimates. The 90% HA reduction on M&Ms comes from the benchmark whose evaluation split was visible during design; how much of it is selection effect is not settled here. Frozen on Prostate158 the same design cuts HA by 39%, and the same template with its own coordinate and depths by 42% on prostate OOD-all (Sec. S10). Under two other objectives the router lowers HA in all six cells of Sec. S12, but most of each reduction is reproduced without the router’s ranking: a matched-quota random control recovers 69%–86% of it, and a different coordinate (n_{reg}) 79%–100%. A reader who treats 39%–42% as the estimate of what the design and its template buy, and 90% as specific to the benchmark it was selected on, reads the evidence as we do.

Baselines, calibration, and scope. The baseline suite is not the published one. Our EATA and CoTTA differ in ways Sec. S1.3 lists, our SAR disables EMA recovery (Sec. S2.4), and MEMO is omitted on ACDC for cost. In Sec. S12, GraTa is ported with recorded deviations and DeYO translated to slices. The routing cut-points are the least stable part of the pipeline: bootstrapping the $n=20$ M&Ms calibration split moves the bucket assignment for a median of 15.2% of evaluation cases, and propagating that perturbation recovers the regime split (Sec. S7). Every threshold presumes a labeled holdout fit once offline, so the label-free claim is about per-case decision time rather than the pipeline, and a site with no labeled data is outside our scope. Estimating the helped/hurt balance without labels, scale-free formulations of the statistic, learning a policy rather than hand-specifying one, and the same keep-or-revert decision on test-time-adapted foundation segmenters are open.

7 CONCLUSION

Episodic TTA conflates a cohort-level budget choice with a per-case one, and cohort means average over the second. Prediction fragmentation reads that decision from two masks, and risky adaptation is easier to identify than beneficial adaptation. Reporting HA beside mean overlap, and a budget *ladder* over a single horizon, is actionable today.

ETHICS STATEMENT

Every dataset used here is public and released by its authors for research use, and the four medical datasets are de-identified: the M&Ms cardiac challenge (Campello et al., 2021), the multi-site prostate benchmark (Liu et al., 2020), whose sites come from the NCI-ISBI 2013 challenge (Bloch et al., 2015; Clark et al., 2013), I2CVB (Lemaître et al., 2015) and PROMISE12 (Litjens et al., 2014), Prostate158 (Adams et al., 2022), RIM-ONE-DL (Batista et al., 2020), Cityscapes (Cordts et al., 2016) and ACDC (Sakaridis et al., 2021). We collected no new human-subject data, accessed no protected health information, and made no attempt to re-identify any subject; each dataset was used within its published licence and intended use.

Two limits on how the results should be read. First, HD95 enters this paper only as a construct-validity check—evidence that HA tracks a boundary metric used in clinical segmentation evaluation (Sec. S1.10)—and not as evidence of clinical performance. Second, the controllers we study are a reliability layer around test-time adaptation, evaluated retrospectively on research benchmarks; nothing here constitutes validation for clinical deployment, which would require prospective evaluation, site-specific calibration and regulatory clearance. We would also flag the failure mode itself as the safety-relevant finding: on the cardiac benchmark a cohort-mean Δ Dice indistinguishable from zero conceals harm to more than half the cases, so a deployment monitored only by mean overlap can degrade individual predictions without any aggregate signal that it is doing so.

AI USE STATEMENT

We used generative AI tools—a code-assistant LLM operated interactively by the authors—for the tasks listed below. The tools helped *implement methods* and analysis code: they drafted and, under the authors’ direction, ran the scripts that re-ran one adaptation pipeline on GPU to regenerate the per-voxel probabilities the certification section needs (Sec. S2.5, Sec. S9); the authors specified the acceptance criteria and verified every output against the archived run. They also drafted the scripts that produce several figures. They *provided feedback on methodology and on experiments*: they proposed the layered acceptance criteria used to validate that rerun (step-1 gradient-norm agreement to 10^{-6} ; cohort-level Dice and HA tolerances; a pre-specified failure disposition fixed before any result was computed), and they argued for and against options that the authors then decided, including whether to re-run that pipeline at all rather than only disclose its provenance. They assisted in *interpreting results*: cross-checking reported numbers against archived outputs and source code, and identifying inconsistencies between text, tables and figures. We did not use generative AI tools to generate synthetic datasets, to develop theoretical models or conceptual frameworks, to formulate or prove mathematical claims, to propose or refine the hypotheses of this work, to clean or reformat datasets, or for translation; qualitative and thematic data analysis is not applicable to this work.

We also used them for creating and editing software code, creating and modifying the figures of this paper, drafting parts of the paper and editing it for readability, editing the $\text{\LaTeX}$ source directly, estimating typeset length, checking terminology and notation for consistency, identifying and formatting references, searching for information, and proposing keywords and a one-sentence summary.

The research questions, the experimental design, the benchmarks and the claims of this paper were determined by the authors. Every change entered the paper only after the authors reviewed and accepted it, and a substantial number of AI-proposed changes were rejected or corrected on inspection, among them an incorrect table number asserted from memory rather than from the compiled cross-references, a table convention that four of the paper’s own tables contradicted. We also solicited several rounds of review from general-purpose AI assistants before submission: one batch contained fabricated references and was discarded; corrections adopted from the others, to scope statements, disclosures, figure encoding and wording, were each verified against the source before entry. We have reviewed all AI-assisted work, and we take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

Every number in this paper comes from a frozen source model, and the appendix is organized so that each can be traced to its source. Most come from archived per-step trajectories; the exceptions

are identified where they appear; these include the certification operating points (Sec. S2.5), the verification-only cardiac signal columns (Sec. S9), the GraTa-cos row (Sec. S9) and the prostate OOD-hard ablation rows (Table S5). Sec. S1 gives the compute environment, software versions, the source models and the episodic protocol, including the learning rates and budgets behind every table. Sec. S1.7 states what is calibrated, on which split, and with which quantile convention—which differs between the M&Ms and driving routers and must be copied per domain rather than assumed. Sec. S10 gives the controller in pseudocode together with the decision-time cost of every signal we compare, and documents the provenance of the design choices that are not derived, including four made with an evaluation split visible. Sec. S1.9 fixes the statistical protocol and Sec. S2.8 is the single authoritative source for every controller-versus-budget interval quoted anywhere in the paper, apart from the SAR probe and the frozen-design test, which are reported in place; no number has two sources.

Three appendix sections exist specifically so that a reader can check rather than trust. Sec. S9 reports an independent rerun of the cardiac trajectories: it reproduces the archived run’s step-1 gradient norms to float precision, its per-step cohort Dice to 3.6×10^{-5} , and it establishes that a cohort HA mean is reproducible across hardware only to about ± 0.002 —a figure we then apply to our own claims, one of which we mark as too small to read as an ordering. Sec. S3 reports pre-specified analysis plans alongside their outcomes, including one expectation that failed because the specification, not the execution, was wrong. Sec. S7 propagates calibration-set perturbation all the way to deployed metrics rather than stopping at routing labels.

All implementation code, dataset configurations, aggregated per-region statistical features, and the archived per-step trajectories will be released on publication; no raw or reconstructed image files from any benchmark are redistributed. The trajectories are the input to every offline analysis reported here; analyses that need per-voxel probabilities or intermediate masks are computed on the verification rerun of Sec. S9 and are identified where reported. None of the control experiments in Secs. S3–S7 requires a GPU or a re-run of adaptation.

REFERENCES

- Lisa C. Adams, Marcus R. Makowski, Günther Engel, Maximilian Rattunde, Felix Busch, Patrick Asbach, Stefan M. Niehues, Shankeeth Vinayahalingam, Bram van Ginneken, Geert Litjens, and Keno K. Bresslem. Prostate158 – an expert-annotated 3T MRI dataset and algorithm for prostate cancer detection. *Computers in Biology and Medicine*, 148:105817, 2022.
- Motasem Alfarra, Hani Itani, Alejandro Pardo, Shyma Yaser Alhuwaider, Merey Ramazanova, Juan Camilo Perez, Zhipeng Cai, Matthias Müller, and Bernard Ghanem. Evaluation of test-time adaptation under computational time constraints. In *International Conference on Machine Learning (ICML)*, 2024.
- Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024.
- Benoît Audelan and Hervé Delingette. Unsupervised quality control of segmentations based on a smoothness and intensity probabilistic model. *Medical Image Analysis*, 68:101895, 2021.
- Christina Baek, Yiding Jiang, Aditi Raghunathan, and J Zico Kolter. Agreement-on-the-line: Predicting the performance of neural networks under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Francisco José Fumero Batista, Tinguaro Diaz-Aleman, Jose Sigut, Silvia Alayon, Rafael Arnay, and Denisse Angel-Pereira. RIM-ONE DL: A unified retinal image database for assessing glaucoma using deep learning. *Image Analysis & Stereology*, 39(3):161–167, 2020.
- N. Bloch, A. Madabhushi, H. Huisman, J. Freymann, J. Kirby, M. Grauer, A. Enquobahrie, C. Jaffe, L. Clarke, and K. Farahani. NCI-ISBI 2013 challenge: Automated segmentation of prostate structures. The Cancer Imaging Archive, 2015.
- Víctor M. Campello, Polyxeni Gkontra, Cristian Izquierdo, Carlos Martín-Isla, et al. Multi-centre, multi-vendor and multi-disease cardiac segmentation: The M&Ms challenge. *IEEE Transactions on Medical Imaging*, 40(12):3543–3554, 2021.

- Ziyang Chen, Yiwen Ye, Yongsheng Pan, and Yong Xia. Gradient alignment improves test-time adaptation for medical image segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 2429–2437, 2025. doi: 10.1609/aaai.v39i3.32244.
- Bowen Cheng, Ross Girshick, Piotr Dollár, Alexander C. Berg, and Alexander Kirillov. Boundary IoU: Improving object-centric image segmentation evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- K. Clark, B. Vendt, K. Smith, J. Freymann, J. Kirby, P. Koppel, S. Moore, S. Phillips, D. Maffitt, M. Pringle, L. Tarbox, and F. Prior. The Cancer Imaging Archive (TCIA): Maintaining and operating a public information repository. *Journal of Digital Imaging*, 26(6):1045–1057, 2013. doi: 10.1007/s10278-013-9622-7.
- Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- Alex Graves. Adaptive computation time for recurrent neural networks, 2016. arXiv:1603.08983.
- Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.
- Jonghyun Lee, Dahuin Jung, Saehyung Lee, Junsung Park, Juhyeon Shin, Uiwon Hwang, and Sungroh Yoon. Entropy is not enough for test-time adaptation: From the perspective of disentangled factors. In *International Conference on Learning Representations (ICLR)*, 2024a. URL <https://openreview.net/forum?id=9w3iw8wDuE>.
- Taeckyoung Lee, Sorn Chottanurak, Taesik Gong, and Sung-Ju Lee. AETTA: Label-free accuracy estimation for test-time adaptation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 28643–28652, 2024b.
- G. Lemaître, R. Martí, J. Freixenet, J. C. Vilanova, P. M. Walker, and F. Meriaudeau. Computer-aided detection and diagnosis for prostate cancer based on mono and multi-parametric MRI: A review. *Computers in Biology and Medicine*, 60:8–31, 2015.
- G. Litjens, R. Toth, W. van de Ven, C. Hoeks, S. Kerkstra, B. van Ginneken, et al. Evaluation of prostate segmentation algorithms for MRI: The PROMISE12 challenge. *Medical Image Analysis*, 18(2):359–373, 2014. doi: 10.1016/j.media.2013.12.002.
- Quande Liu, Qi Dou, Lequan Yu, and Pheng-Ann Heng. Shape-aware meta-learning for generalizing prostate MRI segmentation to unseen domains. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2020.
- Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, Ben Glocker, Fabian Isensee, Jens Kleesiek, Michal Kozubek, et al. Metrics reloaded: Recommendations for image analysis validation. *Nature Methods*, 21(2):195–212, 2024. doi: 10.1038/s41592-023-02151-z.
- Stanislav Nikolov, Sam Blackwell, Alexei Zverovitch, Ruheena Mendes, Michelle Livne, Jeffrey De Fauw, Clemens Meyer, Harry Askham, Bernardino Romera-Paredes, Christopher Chu, et al. Clinically applicable segmentation of head and neck anatomy for radiotherapy: Deep learning algorithm development and validation study. *Journal of Medical Internet Research*, 23(7):e26151, 2021. doi: 10.2196/26151.
- Shuaicheng Niu, Jiaxiang Wu, Yifan Zhang, Yaofo Chen, Shijian Zheng, Peilin Zhao, and Mingkui Tan. Efficient test-time model adaptation without forgetting. In *International Conference on Machine Learning (ICML)*, 2022.
- Shuaicheng Niu, Jiaxiang Wu, Yifan Zhang, Zhiquan Wen, Yaofo Chen, Peilin Zhao, and Mingkui Tan. Towards stable test-time adaptation in dynamic wild world. In *International Conference on Learning Representations (ICLR)*, 2023.

- Shuaicheng Niu, Chunyan Miao, Guohao Chen, Pengcheng Wu, and Peilin Zhao. Test-time model adaptation with only forward passes. In *International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 38298–38315. PMLR, 2024. URL <https://proceedings.mlr.press/v235/niu24a.html>.
- Christos Sakaridis, Dengxin Dai, and Luc Van Gool. ACDC: The adverse conditions dataset with correspondences for semantic driving scene understanding. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- Mona Schirmer, Metod Jazbec, Christian A. Naesseth, and Eric Nalisnick. Monitoring risks in test-time adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Steffen Schneider, Evgenia Rusak, Luisa Eck, Oliver Bringmann, Wieland Brendel, and Matthias Bethge. Improving robustness against common corruptions by covariate shift adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *International Conference on Machine Learning (ICML)*, 2020.
- Surat Teerapittayanon, Bradley McDanel, and H. T. Kung. BranchyNet: Fast inference via early exiting from deep neural networks. In *International Conference on Pattern Recognition (ICPR)*, 2016.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations (ICLR)*, 2021.
- Qin Wang, Olga Fink, Luc Van Gool, and Dengxin Dai. Continual test-time domain adaptation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Marvin Zhang, Sergey Levine, and Chelsea Finn. MEMO: Test time robustness via adaptation and augmentation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Yubo Zhou, Jianghao Wu, Wenjun Liao, Shichuan Zhang, Shaoting Zhang, and Guotai Wang. TEGDA: Test-time evaluation-guided dynamic adaptation for medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, volume 15965 of *Lecture Notes in Computer Science*, pp. 628–637. Springer, 2025.

S1 REPRODUCIBILITY DETAILS

S1.1 COMPUTE AND SOFTWARE

All experiments run on one NVIDIA RTX 6000 48GB; no cluster or machine identifiers are given here. Software: Python 3.11.15, PyTorch 2.6.0, CUDA 12.4, MONAI 1.3.2, nnU-Net v2.2.1. Source models are frozen: no offline or persistent target-domain training is performed, and no adapted weights survive a case (Sec. S1.3); “frozen” refers to the source checkpoint, not to an absence of gradient steps. Code and configuration files will be released on publication.

S1.2 SOURCE MODELS (M_0)

- *Cardiac / prostate MRI*: frozen nnU-Net source model (InstanceNorm backbone); no retraining.
- *Driving (ACDC)*: frozen Cityscapes-pretrained SegFormer-b0 (19 classes); no retraining. Adaptation updates the affine parameters of the encoder `LayerNorm` modules only (30 modules, 7,168 parameters, 0.19% of the network); the single `BatchNorm` in the decode head is not updated.

S1.3 EPISODIC TTA PROTOCOL

Per case: reset to source weights M_0 , adapt for a fixed budget of K steps, then reset. Adaptation objective: entropy minimization on normalization affine parameters (TENT-style; EATA adds entropy filtering with $e_{\text{margin}}=0.45$ and an anti-forgetting anchor with weight 1.0). The margin never binds on the medical benchmarks: at M_0 all 2,434 cardiac evaluation slices and all 1,420 slices of the 124-case prostate pool, a superset of the evaluation split, pass it, the largest slice entropy observed being $22\times$ below it, so those rows are entropy-filtered in configuration only and are effectively unfiltered in effect. The anchor is an *unweighted* ℓ_2 penalty toward the source weights, $\sum_p \|p - p_0\|^2$, not the Fisher-weighted form of the original EATA; we keep the symbol λ_{Fisher} for the configuration key but the term it multiplies carries no Fisher information matrix. This differs from the published EATA regularizer, and uniform anchoring was used throughout, so our EATA rows should be read as an entropy-filtered, source-anchored baseline rather than as a reproduction of that method. Baseline configurations: MEMO uses 4 augmentations with consistency weight 1.0; our episodic CoTTA variant uses 2 light augmentations, teacher EMA decay 0.999, stochastic restore probability 0.01 and consistency weight 10.0. Optimizer: Adam (default β , no weight decay), learning rate as below, batch size 1 (episodic, one case at a time). Main tables use a deterministic episodic reset; CRC / multi-seed stability sweeps use seeds $\{0, \dots, 4\}$ where reported.

Learning rates / budgets used.

- Medical (prostate/cardiac): $\text{lr } 5 \times 10^{-4}$, $K=4$ (fixed-4 baseline); controllers adapt within the same $K=4$ budget (mean steps <4). Main-table fixed-1/2/3 are the step- k checkpoints of the same episodic EATA fixed-4 trajectory (no separate re-run).
- ACDC mild budget: $\text{lr } 10^{-4}$, $K=4$.
- ACDC aggressive budget: $\text{lr } 2 \times 10^{-3}$, $K=8$.
- Budget-axis sweep (Table S6, Sec. S2.2): $\text{lr} \in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$.

S1.4 BUDGET SELECTION AND FIXED- k LADDERS

This section holds the full budget-axis evidence behind the main-text ladders: the holdout $\rightarrow$ eval transfer experiments on both medical splits, and the complete eight-step ACDC rain ladder that the main text abridges to $k \in \{1, 2, 4, 8\}$.

Fixed- k holdout $\rightarrow$ eval transfer (prostate OOD-all). Using the same per-site 20% holdout as the main tables ($n_{\text{holdout}}=23$, $n_{\text{eval}}=93$) and the EATA fixed-4 trajectory checkpoints:

- Holdout mean HA by k : 0.148 / 0.169 / 0.206 / 0.287 for $k=1..4$ (argmin $k=1$).

- Eval mean HA by k : 0.242 / 0.211 / 0.171 / 0.262 (argmin $k=3$).
- Every standard selection rule on the holdout (min HA, max Dice, HA−0.5Dice, HA−Dice) picks $k=1$; none recovers the eval optimum.
- Deploying holdout-selected $k=1$ on eval incurs HA regret +0.070 vs. the eval oracle $k=3$.
- Case bootstrap ($B=5000$): $P(\text{holdout selects } k=1)=0.68$, $P(\text{selects } k=3)=0.07$; joint resampling gives $P(k_{\text{holdout}}=k_{\text{eval}})=0.071$, with mass on the pair (1, 3).
- Per-site argmin HA agrees on only 3/6 sites between holdout and eval.

Paired bootstrap of the controllers against these budgets ($B=5000$, seed 1000) is collected in Sec. S2.8, which states its own scope.

Fixed- k holdout→eval transfer (M&Ms vendor B). Holdout is the official vendor-B calib split ($n=20$; `vendorB_calib.txt`); eval is B-test ($n=230$). Overlap is empty. Holdout uses a fresh EATA fixed-4 run on B-calib; eval reuses the archived B-test fixed-4 trajectory checkpoints:

- Holdout mean HA by k : 0.084 / 0.046 / 0.070 / 0.221 for $k=1..4$ (argmin $k=2$).
- Eval mean HA by k : 0.186 / 0.164 / 0.129 / 0.177 (argmin $k=3$).
- Eval mean Dice by k : 0.8506 / 0.8504 / 0.8497 / 0.8490 (spread 0.002, the Dice range quoted in Table 1’s caption).
- Min-HA, max-Dice, and HA−0.5Dice on the holdout all pick $k=2$; none recovers the eval optimum $k=3$.
- Deploying holdout-selected $k=2$ on eval incurs HA regret +0.035 vs. the eval oracle $k=3$.
- Case bootstrap ($B=5000$, seed 1000): $P(\text{holdout selects } k=2)=0.64$, $P(\text{selects } k=1)=0.36$, $P(\text{selects } k=3)=0$; joint resampling gives $P(k_{\text{holdout}}=k_{\text{eval}})=0.003$, with mass on the pair (2, 3).

Paired bootstrap on B-test is in Sec. S2.8.

Fixed- k ladder on ACDC rain (TENT, $\text{lr } 2 \times 10^{-3}$). Extracted from the archived val trajectory checkpoints ($n=100$), identical to the TENT₈ run the main text reports:

- Mean HA by $k=1..8$: 0.264 / 0.327 / 0.358 / 0.385 / 0.408 / 0.422 / 0.439 / 0.448 (strictly monotone; argmin $k=1$).
- Mean mIoU: 0.378 / 0.385 / 0.387 / 0.389 / 0.389 / 0.389 / 0.390 / 0.387 (peaks at $k=7$, dips at $k=8$).
- Train→val min-HA selection of k : both pick $k=1$; bootstrap $P(k_{\text{train}}=k_{\text{val}})=1.0$ ($B=5000$; $n_{\text{train}}=400$).

Controller-vs-budget intervals on this ladder are in Sec. S2.8.

S1.5 FRAGMENTATION SIGNAL

Computed from predicted masks only (no labels, no gradients at decision time): number of disagreement connected components n_{reg} , disagreement ratio δ , their step deltas, and mean predictive entropy on disagreement voxels. Connected components use `scipy.ndimage.label` default face-connectivity (4-connected in 2D; 6-connected in 3D); minimum region size 16 voxels (medical) / 200 px (ACDC). Mask convention: on the medical benchmarks M_0 and M_k are binarized to foreground (any positive label) before the disagreement mask is formed, so only foreground/background edits count; on ACDC the 19-class label maps are compared directly, so any label change counts. HA (Sec. 3) uses the same masks, with GT binarized the same way on the medical benchmarks.

S1.6 SIGNAL–HARM ASSOCIATION ACROSS DOMAINS

The cross-benchmark rows of main-text Table 2 use the protocol stated there: Spearman correlations pooled over (case, step) pairs with step ≥ 1 (step 0 is excluded, since HA=0 there by construction),

Domain	Coordinate	Low	Mid	Hard
Prostate	n_{reg}	depth 1	depth 1	rollback
M&Ms	δ	depth 3	depth 2, then shrink ($\alpha=0.5$)	rollback
ACDC	δ	stop rule, cap 8	stop rule, cap 4	rollback

Table S1: Routing coordinate and per-bucket action by domain. Cut-points are tertile percentiles of the step-1 coordinate on the calibration split, with the exact convention differing between the M&Ms and driving pipelines (Sec. S7); prostate: per-site 20% OOD holdout, $n=23$; M&Ms: vendor-B calib, $n=20$; ACDC: per-condition train. The two medical routers deploy a fixed depth per bucket, so low and mid coincide on prostate and differ by one step and the shrink action on M&Ms; the driving router runs the parameter-free Δn_k stop rule of Sec. 4.2 within a per-bucket cap (the calibrated τ -stop is a separate controller), and its depth therefore varies by case (Table S2).

Domain	Bucket	n	std	range
M&Ms	low / mid	54 / 46	0.00 / 0.00	3 / 2
Prostate	low / mid	27 / 24	0.00 / 0.00	1 / 1
ACDC fog	low / mid	52 / 30	0.64 / 0.66	1–5 / 1–4
ACDC rain	low / mid	38 / 41	0.37 / 0.64	1–2 / 1–4
ACDC snow	low / mid	31 / 40	0.77 / 0.99	1–4 / 1–4
ACDC night	low / mid	22 / 41	2.56 / 1.30	1–8 / 1–4

Table S2: Within-bucket dispersion of deployed depth. The four medical buckets are single-valued; all eight driving buckets are not. This is the measurement behind the distinction drawn in Table S1: on the medical benchmarks the fragmentation state enters the bucket assignment and not the depth. The dispersion is largest on night, where M_0 is near-collapsed and Δn_k has no stable rise point (Sec. 5.3).

and a Δ column that differences consecutive steps within a case. Apart from the verification-rerun rows marked in its caption and the GraTa-cos row (Sec. S9), no new runs were performed for it; every other value is computed from the per-step statistics already stored for the main-table trajectories. The same statistics give the signal–signal correlations the main text quotes for complementarity: on prostate, over the same step- ≥ 1 pairs, n_{reg} correlates $+0.034$ with mean entropy and -0.101 with mean MSP (δ : $+0.035$ and -0.113). Clustered intervals, the per-condition and per-step breakdowns, and the attribution of the aggressive-budget collapse to update size rather than trajectory depth are all in Sec. S4, which is the single source for those numbers.

S1.7 CALIBRATION PROTOCOL

z-score statistics and the stop threshold τ are fit *once* on a held-out labeled split (per-site for medical, ACDC train split for driving; sizes in Table S1), disjoint from the online test stream. Which statistics are needed depends on the controller. FragSubset scores with the two-term risk of Eq. 2 and therefore fits both pairs on the medical benchmarks; the ACDC τ -stop scores with $r_k = z(n_k)$ alone, so $(\mu_\delta, \sigma_\delta)$ do not enter there. The routers use no z-scores at all: they split a raw per-domain coordinate at tertiles of the calibration split (Table S1). FragStop is parameter-free. Fitted values: ACDC full-train $\mu_n=20.1$, $\sigma_n=6.8$, $\tau=-0.16$; prostate holdout $\mu_n=14.87$, $\sigma_n=21.20$, $\mu_\delta=0.0068$, $\sigma_\delta=0.0113$; M&Ms calibration split $\mu_n=0.70$, $\sigma_n=0.78$, $\mu_\delta=0.00028$, $\sigma_\delta=0.00020$. No labels or gradients are used at decision time.

COQR router. Case-level online quantile routing makes only the routing decision at test time, from tertile thresholds of the fragmentation score. On prostate OOD-all ($n=93$) it routes 27 cases low, 24 mid and 42 hard: low and mid cases each deploy at depth 1 by bucket policy, not by a stop-rule trigger (only 3 of the 51 satisfy $\Delta n_2 > 0$), and hard cases are rolled back to M_0 . The hard bucket is also where the source model is weakest (mean M_0 Dice 0.578, against 0.869 in the two adapting buckets). Because the cut-points are percentiles of the *calibration split*, the bucket shares on the evaluation set need not be even, and on both medical benchmarks they are not: 45.2% of prostate cases and 56.5% of M&Ms cases land in the hard bucket. A near-even split would in fact be the suspicious outcome, since it is what fitting the percentiles on the evaluation pool itself would produce.

What decides whether the per-case decision is worthwhile is not how evenly the buckets fill but the helped/hurt composition of the retained bucket (main text). We report the router on OOD-all rather than on the $n=34$ OOD-hard subset because the deployed cut-points put 28 of the 34 in the hard bucket and leave the low bucket empty, so there are no per-bucket means to report. On ACDC the router uses per-condition cut-points, since shift severity differs sharply across conditions; low and mid bucket sizes are in Table S2, the hard buckets hold 18 (fog), 21 (rain), 29 (snow) and 43 (night) cases, and the actions are in Table S1.

Conformal gate. $\hat{\lambda}$ is selected on the calibration split via mean-risk conformal risk control, with a fixed-denominator surrogate.

Routing coordinate by domain. Table S1 gives the instantiation. This qualifies what “transfers” means: the M&Ms→ACDC port keeps the routing coordinate, refits the tertiles and replaces the fixed depths with the stop rule under per-bucket caps, whereas the prostate router uses a different coordinate of the same fragmentation state s_k .

What “seed-only” means. On M&Ms and ACDC the cut-points are percentiles of the step-1 scores of the calibration cases only, 33.33/66.67 on M&Ms and 33/66 on ACDC (*seed-only*: the seed pool is never extended with test cases and no online sliding update is applied). The implementation also offers a *hybrid* mode that appends each observed test case to the pool and refits the percentiles over a sliding window; we do not use it, because it would make the thresholds a function of the test stream. Seed-only is the setting reported everywhere in this paper. On ACDC the cut-points are fitted per condition, which assumes the condition is known at deployment; a single pooled threshold would be the honest alternative for a system that must infer it, and we did not run it. The pools are the calibration splits of Table S1; on prostate the step-1 n_{reg} tertiles are 4.0 and 10.7.

Stability of the cut-points. Because the buckets are fixed offline, their sensitivity to the calibration sample matters, and that sample is small. Bootstrapping the step-1 δ pool of the $n_{\text{calib}}=20$ M&Ms calibration split (5000 resamples) gives $\tau_{\text{lo}} = 0.000169$ [0.000107, 0.000216] and $\tau_{\text{hi}} = 0.000263$ [0.000191, 0.000559], and the induced bucket assignment on the evaluation set changes for a median of 15.2% of cases (upper tail $\approx 38\%$). The cut-points are therefore the least stable part of the controller. That is the price of calibrating on twenty cases, and it is a price we pay deliberately: the only larger pool available is the evaluation stream itself, which seed-only calibration rules out. We would rather report a wide resampling interval than a narrow one obtained that way. The same constraint bounds the certified gate at small n_{cal} (Sec. S2.5), so calibration-set size is a real limit on this pipeline rather than an artifact of one experiment. The consequences of that perturbation for deployed Dice and HA are propagated in Sec. S7; neither analysis re-runs adaptation under new seeds.

How steps are counted. Across all domains the reported *Steps* column is the deployed depth of the state actually returned: a case routed hard is probed for one step, reset to M_0 and recorded as 0 steps (`rolled_back=1`), while low and mid cases are recorded at the step where they stop. One qualification: on M&Ms the mid bucket shrinks rather than merely halting, so its 46 cases deploy the weights interpolated half-way back to M_0 ($\alpha=0.5$) after their second step. That state is not any M_k checkpoint, and we record it at the step it was taken from (2). The *Steps* column is therefore the depth along the trajectory at which the deployed state was formed, not a count of updates surviving in the returned weights; for every method except the M&Ms mid bucket the two coincide. The $|D|$ used for its HA is computed against the prediction actually deployed from the interpolated weights, consistent with the M_0 -referenced convention of the main text. The definition is identical in the M&Ms and driving pipelines. Differences between domains therefore reflect the policies attached to each bucket, not different accounting: on M&Ms the mean of 1.10 comes from 130 hard cases at 0 steps, 46 mid at 2 and 54 low at 3; on prostate OOD-all the mean of 0.55 comes from 42 hard at 0 steps and 51 low or mid cases that each stop at 1; on ACDC the per-condition means are 0.96 (fog), 0.97 (rain), 1.01 (snow) and 1.67 (night), the last being higher because low and mid cases there run longer (3.23 and 2.59 steps) before the rule halts them. The τ -stop controller uses a different convention and is not comparable on this column: it never rolls back and is recorded with a floor of one step, so its rain mean is 2.49 rather than ~ 1 .

S1.8 BASELINES

The four stop rules compared in Sec. S5 share one parameter-free form, halting at the first step at which the statistic worsens relative to the previous step, and differ only in the statistic (n_{reg} rising, mean entropy rising, MSP falling, gradient norm rising). None of them uses a calibrated threshold, so that comparison is controlled by construction; the calibrated τ -stop of the ACDC section is a separate controller. As adaptation objectives we wrap TENT, EATA (fixed- K), MEMO, and an episodic light-augmentation CoTTA variant ($n_{\text{aug}}=2$), all under the same episodic reset and the same M_0 -referenced HA. GraTa’s alignment enters Table 2 only as a read-only signal; the full objective is in Sec. S12.

S1.9 EVALUATION AND STATISTICS

Split sizes: prostate OOD-hard $n=34$, OOD-all $n=93$, pooled trajectories $n=124$ (the 93 evaluation cases plus the 23 per-site holdout cases and 8 source-domain G-val cases, i.e. every case with a stored EATA trajectory; used only for the signal–HA correlations, never for a deployment number), HA–HD95 cohort $n=34$ (the OOD-hard split). The prostate OOD-hard *pool* is exactly sites B and E ($30+12=42$ cases); the 34 of them that fall in the evaluation split are the OOD-hard split reported in the tables, and the remaining 8 B/E cases fall in the holdout (a different set from the 8 G-val cases above). We call the full 42 the *B/E signal-analysis pool*: it is used only for descriptive signal analysis (the region-geometry contrast of Sec. S1.11 is drawn from these sites), never to fit a deployed threshold. All deployed thresholds come from the per-site 20% holdout (Table S1). M&Ms vendor-B $n=230$; ACDC val per condition $n=100$ for fog/rain/snow and $n=106$ for night (the dataset’s own split). Every method in the prostate OOD-all panel is run on the identical case set, so the router, the fixed-budget baseline and the external baselines are directly comparable there.

Pooling conventions. Two distinct senses of “pooled” appear in this paper and are unrelated. Figure 1(A) reports mIoU micro-averaged over all pixels of the condition, which puts M_0 at 0.385 on ACDC rain, whereas every table reports the per-image mean, which puts it at 0.364 (Table S7); the panel is there for the shape of the two curves, and its y values are not comparable with the tables. Elsewhere “pooled” refers to case sets (the $n=124$ prostate signal-analysis pool above) and to correlations pooled over (case, step) pairs, neither of which concerns pixel averaging.

Table conventions. Bold marks the best value among those compared within a block (highest for Dice, mIoU, ρ and AUROC, lowest for HA and deployed steps) and is omitted where the text reports no separation. Interval tables carry no bold: the comparison is made by the intervals themselves. Structural bold (row names, algorithm keywords) is not a comparison. One table departs from the rule and says so in its caption: Table 5 bolds the larger of helped and hurt on each router row, which on prostate is the worse of the two.

Within the deployed pipeline, labels are used offline in exactly four places: (1) to select the reference budget on the holdout; (2) to fit the stop threshold τ on the calibration split (a quantile of the fragmentation risk at the *oracle* early-stop step, which needs the metric); (3) to fit $\hat{\lambda}$ for the conformal gate; and (4) to score HA and Dice / mIoU after the fact; the design choices made with an evaluation split visible are a separate exposure, disclosed in Sec. S10. The fragmentation statistics themselves, (μ_n, σ_n) , $(\mu_\delta, \sigma_\delta)$ and the routing cut-points, are computed from predicted masks on those same splits and use no labels. Nothing of any kind is fit inside the adaptation loop or the per-case decision. Paired bootstrap with 5000 resamples for all confidence intervals; the tier labels A / A' / B / C used throughout are the ones defined in the main-text footnote (deployable budget / default fixed-4 / retrospective eval-best k / descriptive); the Tier-C positioning is Fig. S3. Read panel by panel: our controllers move to the safer side of *their own budget reference* in every panel, and on ACDC the ordering TENT $\rightarrow$ τ -stop $\rightarrow$ router moves rightwards on all conditions (fog is not shown; Table S7); on prostate OOD-hard MEMO and CoTTA sit safer than our controllers, at the accuracy cost documented in the main text; on prostate OOD-all the router is the safest point shown but sits on the M_0 Dice line, the boundary Sec. 5.2 discusses and Table 5 decomposes. Holm correction is applied in exactly one place: the ordering claim *among our own controllers* (modulation beyond stopping, FragSubset vs. FragStop on prostate OOD-hard). Comparisons against the retrospective budget are reported as single paired-bootstrap intervals without correction, because each is a pre-specified comparison against one named reference rather than a member of a searched family. The ACDC intervals and $P(\Delta < 0)$

values are computed per comparison from the paired bootstrap and are reported *uncorrected*; with four conditions and two reference methods they should be read as describing each condition rather than as a family-wise test. We flag this because the ACDC HA reductions are large and consistent enough that no correction would change their sign, whereas the small mIoU differences there are exactly the quantities a correction would attenuate.

S1.10 HA–HD95 CONSTRUCT-VALIDITY CHECK

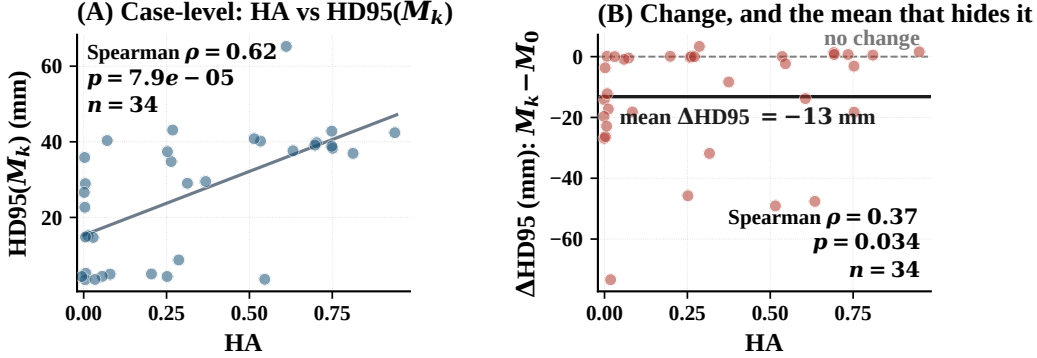


Figure S1: **HA vs. a clinically used boundary metric** (prostate OOD-hard, EATA fixed-4, $n=34$; lower is better everywhere). (A) vs. HD95(M_k), $\rho=0.62$; the straight line is an ordinary least-squares fit shown as a guide. (B) vs. ΔHD95 ($\rho=0.37$): black=cohort mean (-13 mm), dashed=no change; the mean improves while HA accumulates.

Boundary metric: HD95 (medpy 0.5.2) on the same masks used for HA (prostate OOD-hard, EATA fixed-4, $n=34$). The fixed-4 mask dump covers 50 cases (all of sites B and E plus eight source-site G cases), and we restrict the check to the 34 that are in the OOD-hard evaluation split, so the cohort matches the split reported in the prostate tables. Spearman $\rho(\text{HA}, \text{HD95}_{M_k})=0.62$ ($p=7.9\times 10^{-5}$); $\rho(\text{HA}, \Delta\text{HD95})=0.37$ ($p=0.034$); cohort-mean $\Delta\text{HD95}=-13$ mm.

The same check on cardiac uses the Sec. S9 rerun masks (all 230 vendor-B evaluation cases, EATA fixed-4, per-case spacing from the image headers, foreground-binarized as HA is). HA tracks the *change* in boundary error at $\rho=0.34$ ($p=1.0\times 10^{-7}$) and the adapted boundary error itself at $\rho=0.27$ ($p=4.6\times 10^{-5}$), against 0.37 and 0.62 on prostate. The weaker level correlation is what the sparsity of cardiac edits predicts: HA is zero on 144 of the 230 cases and HD95 does not move on 173, so most of the cohort is tied at the origin; among the 30 cases where both quantities move, $\rho(\text{HA}, \Delta\text{HD95})$ is 0.78. The level correlation is weaker because the edit set is smaller: at fixed-4 the cardiac disagreement covers $\delta=3.3\times 10^{-4}$ against 5.6×10^{-3} on prostate (Sec. S6), so HA sits at zero on most cardiac cases and cannot order their absolute boundary error. The dissociation also takes a different form. On prostate the cohort mean improves by 13 mm while HA accumulates; on cardiac it barely moves (-0.60 mm, median 0) while HA still has per-case structure. That is the shape of Sec. 3.1's reading of ΔDice on this benchmark, where the cohort mean is indistinguishable from zero while 58.7% of cases are individually worse: on cardiac both a region metric and a boundary metric average to nothing over a cohort that contains harm.

S1.11 REGION-LEVEL GEOMETRY OF HARMFUL VS. BENEFICIAL EDITS

The main text claims fragmentation as a *case-level* signal and explicitly does not claim that an individual harmful edit is geometrically identifiable. This section is the evidence for that disclaimer. On the same EATA fixed-4 dumped masks we label every disagreement component as harmful or beneficial by the HA definition of the main text (identical connectivity and $\tau_{\text{area}}=16$; neutral components, where M_k and M_0 are correct on equally many voxels, are 0.1% of regions on prostate and 0.4% on M&Ms and are dropped) and test the natural region-level reading of the signal: that harmful edits are smaller, less compact, more often tiny islands, and more detached from the bulk of the disagreement.

Feature	mean H	mean B	p	Cliff's d
<i>Prostate OOD-hard</i> ($n=34$ cases; 218 H, 524 B)				
area (vox)	614.7	405.5	0.054	−0.09
compactness A/S^2	0.0279	0.0246	0.053	+0.09
dist. to largest	50.7	89.5	8×10^{-23}	−0.46
frac. of $ D $	0.049	0.031	4×10^{-7}	+0.24
small @5%	0.789	0.903	3×10^{-5}	−0.11
<i>M&Ms vendor-B</i> ($n=230$ cases; 144 H, 139 B)				
area (vox)	159.4	165.5	0.94	−0.00
compactness A/S^2	0.0251	0.0248	0.94	+0.00
dist. to largest	14.7	19.7	0.031	−0.14
frac. of $ D $	0.288	0.246	0.15	+0.10
small @5%	0.153	0.209	0.22	−0.06

Table S3: Region-level contrast between harmful (H) and beneficial (B) edits, EATA fixed-4. Mann–Whitney U (two-sided); Cliff’s d is $H - B$, so a negative d means harmful regions score *lower*. We write it d rather than δ to keep it distinct from the disagreement ratio δ used throughout. “small @5%” is the fraction of regions below 5% of $|D|$.

That reading is not supported (Table S3). On prostate three contrasts reach significance, and the two large ones run the *opposite* way: harmful regions sit much closer to the largest disagreement component ($d=-0.46$) and take a larger share of $|D|$ ($d=+0.24$), while it is the *beneficial* edits that are more often tiny islands (small@5% 0.90 vs. 0.79; the gap is stable at 2%, 5% and 10% thresholds). Area and compactness do not separate the two classes at $\alpha=0.05$. On M&Ms the contrast is essentially null, with only distance-to-largest weakly echoing prostate ($p=0.031$), so the one prostate effect that is large does not even replicate in strength across datasets. Mechanistically, harmful edits on prostate look like attached over-growth of the main disagreement mass rather than a swarm of debris.

None of this weakens the case-level claim: a rising n_{reg} says the update produced many mutually unrelated changes, which predicts case-level HA (main-text signal–HA table) without requiring that any single harmful region be smaller or more scattered than any single beneficial one. It does rule out a region-level gate built on the same geometry, because an edit-wise filter keyed to “small and detached” would preferentially discard the beneficial edits. That is why our controllers accept or reject at the case level. Per-region features and the plotting code will be released on publication.

S1.12 CASE-LEVEL HELP/HURT DECOMPOSITION

For each case we compute $\Delta = \text{metric}(M_k) - \text{metric}(M_0)$ (Dice on medical, mIoU on ACDC): a case is *rolled back* if the router locked the deployed output to M_0 , and otherwise *helped* ($\Delta > 0$) or *hurt* ($\Delta < 0$). Classification is by decision, not by a numerical tolerance. This matters on prostate, where a rollback re-runs 3D sliding-window inference instead of reusing the cached M_0 and leaves differences of order 5×10^{-5} per voxel; an $|\Delta|$ threshold would mis-count those cases, whereas the routing itself is exact (rolled_back = 1, step 0, HA 0 for all 42 hard cases). Under the semantic definition the prostate rolled-back share (45.2%) matches its hard bucket exactly, and the ACDC and M&Ms figures are unchanged. The uncontrolled reference differs by domain, matching each main table: fixed-4 on M&Ms and prostate, TENT₈ on ACDC. Baselines have no rollback action, so their rolled-back share is 0 by construction.

S1.13 QUALITATIVE ACDC EXAMPLE

Figure S2 is a single-case illustration of the ACDC rain failure mode of Figure 1(A) (mIoU flat or down while HA rises under fixed- K TENT), here at the aggressive budget rather than the mild one shown there. Same protocol as the main-text ACDC rain comparison (lr 2×10^{-3} , $K=8$; full-train τ calibration, Sec. S1.7).

Split	Method	helped ($\mu_{\Delta+}$)	hurt ($\mu_{\Delta-}$)	rolled back	overall μ_{Δ}
ACDC fog	TENT ₈	52.0 (+0.054)	48.0 (−0.034)	—	+0.012
ACDC fog	Router	44.0 (+0.023)	38.0 (−0.009)	18.0	+0.007
ACDC rain	TENT ₈	69.0 (+0.047)	31.0 (−0.031)	—	+0.023
ACDC rain	Router	59.0 (+0.020)	20.0 (−0.012)	21.0	+0.010
ACDC snow	TENT ₈	61.0 (+0.050)	39.0 (−0.033)	—	+0.018
ACDC snow	Router	46.0 (+0.027)	25.0 (−0.009)	29.0	+0.010

Table S4: Case-level decomposition of Δ against M_0 on the three remaining ACDC conditions (percentages of all cases in the condition); M&Ms, prostate and ACDC night are in Table 5. The router lowers the hurt fraction on each of these three as well, and on all three it retains more helped cases than hurt ones. On prostate a further 4.3% of cases adapt but leave the prediction numerically unchanged and fall into none of the three classes; an exactly zero Δ for an adapted case does not occur on M&Ms.

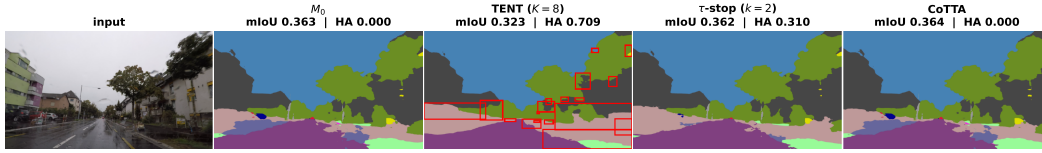


Figure S2: Harm invisible to mIoU (ACDC rain, GP020402_frame_000748). L→R: input, M_0 , TENT ($K=8$; every harmful region above the 200 px threshold is boxed in red), the τ -stop of Sec. 4.2 (ours), CoTTA. TENT *lowers* mIoU (0.363→0.323) while HA explodes to 0.709; τ -stop stops near the mIoU peak ($k=2$), keeping mIoU 0.362 at HA 0.310 (this case’s HA coincides with the rain cohort mean of Table S7 to three decimals, 0.3101 against 0.3103; its mIoU does not); episodic CoTTA barely adapts (output $\approx M_0$, mIoU 0.364). Unlike the medical qualitative rows, whose masks are replayed from the archived dumps, this panel recomputes the TENT, τ -stop and CoTTA predictions for this image, so reproducing it needs a forward pass rather than the stored trajectories.

The archived script for this figure was edited after the published panel was produced; the edit affects only the four panel-title strings, not the case, the calibration constants, the minimum-region size or the metric conventions, all of which we checked line by line against the values quoted above. Re-running it reproduces the panel with slightly different title formatting. We release the script as it stands rather than reconstructing the earlier strings from the rendered figure.

S1.14 CROSS-MODALITY DISAGREEMENT SCALE (FUNDUS VS. CARDIAC)

The main-text Limitations cite a concrete scale gap between fundus and cardiac disagreement ratios. Protocol and numbers:

- *Statistic*. Disagreement ratio $\delta = |D|/|\Omega|$, where $D = \{x : M_0(x) \neq M_k(x)\}$ and $|\Omega|$ is the number of spatial locations (same definition as the main-text fragmentation signal).
- *Fundus*. Episodic TENT on RIM-ONE-DL (Batista et al., 2020) ($n=174$ hospital-test OOD), with a frozen REFUGE-trained source; mean disagree area $\delta \approx 0.0389$ (3.9%) at the adapted state used for the cross-modality scope probes.
- *Cardiac (comparator)*. Episodic EATA / default TTA on M&Ms Vendor-B; cohort-mean step-1 δ is on the order of 10^{-3} (≈ 0.001 ; the measured fixed-4 step-1 mean is 5.45×10^{-4}).
- *Ratio*. Three matched comparisons give values between $37\times$ and $71\times$; we quote the conservative figure of roughly $40\times$. Absolute δ cutoffs therefore do not transfer across this modality shift without recalibration; relative features (Δn_k , within-domain z-scores / tertiles) remain the intended deployment protocol.

What the ported threshold does. A threshold carried over without recalibration ends below its own fixed-budget baseline. The run behind this is episodic TENT on RIM-ONE ($n=174$) with the medical gate applied at its in-domain setting. Un-gated TENT reaches mean Dice 0.646 against the frozen

source’s 0.635. Porting the gate rolls back 97.8% of cases (whole-image) or 85.1% (region-level), which lands at 0.635 and 0.636, at the un-adapted source and below the un-gated run it was meant to protect. The failure mode is exactly what the scale gap predicts: a cut-point calibrated where $\delta \sim 10^{-3}$ rejects almost everything where $\delta \sim 4 \times 10^{-2}$.

This is a *scope boundary*, not a fundus leaderboard claim: we do not report FG-TTA controllers as primary results on fundus.

S1.15 QUALITATIVE CASE SELECTION (MEDICAL FIGURE)

The main-text medical qualitative figure (“Harm invisible to Dice”) illustrates fixed-budget over-adaptation that is nearly invisible to mean overlap. The main text shows the M&Ms row only; the prostate row is Fig. S5 below. Ideal selection would jointly satisfy four desiderata on the *displayed* slice: (i) high HA under fixed-4, (ii) flat $|\Delta \text{Dice}|$, (iii) both helpful and harmful disagreement components visible on that slice, and (iv) $M_0 \approx \text{GT}$ visually. On M&Ms Vendor-B fixed-4 dumps ($n=230$), *no* slice meets all four simultaneously. We therefore prioritize (i)–(ii), since those are the properties the figure claims to illustrate, and take (iii) when it does not conflict with flat ΔDice .

Picks.

- *M&Ms (main text)*: A8E1F4_frame09. Panel metrics are *slice-level*. Fixed-4 raises slice Dice $0.55 \rightarrow 0.56$ while $\text{HA}=0.90$; the display slice contains both helpful and harmful edits. On this slice GT has no RV voxels, so mean FG-Dice is blind to the large false-RV expansion (FP on an empty class does not move Dice); HA still flags it. Weak M_0 is intentional domain-shift motivation. Router returns $\text{HA}=0$ at M_0 .
- *Prostate (Fig. S5)*: prostate_B.0017. High- M_0 complement (slice $D \approx 0.93$); fixed-4 $\text{HA}=0.62$ at near-flat Dice; FragSubset $\text{HA}=0$. The volume mixes helpful and harmful components in 3D (7 H / 8 B); the peak-harm display slice is harm-dominated.

A rejected alternative (B5T6V0_frame00) has high M_0 Dice on the peak-harm slice but drops Dice by ~ 0.13 under fixed-4, which would *contradict* an “invisible to Dice” caption; we do not use it.

The cross-backbone figure of Sec. S11 (Fig. S6) uses a different rule: the same case under both backbones, both M_0 within the matched band, and the median transformer-side disagreement volume among those. Its slice is the one carrying the most nnU-Net supra-threshold disagreement area. That slice rule was fixed after an earlier one (most transformer-side disagreement) tied on the chosen case and, at either tied slice, showed no nnU-Net component.

S2 RESULTS DEFERRED FROM THE MAIN TEXT

This section holds material compressed out of the main paper for space. Every item is referenced from the main text at the point where it was cut.

S2.1 PROSTATE OOD-HARD PANEL

The main text reports prostate MRI on the full OOD-all set ($n=93$) and summarizes this subset in one paragraph. Its cut-points are fitted once on the OOD holdout and applied to the 93-case stream; filtering that stream down to its 34 hardest cases leaves 0/6/28 cases in the three buckets, which is too degenerate for a bucket-level number to mean anything. Any router row a reader reconstructs for this subset is descriptive (Tier C) and should not be compared against the intervals reported for the full split. Table S5 gives the panel. The ordering reverses here: MEMO attains the best HA overall and the best Dice among the non-fixed methods, at the full four-step budget with extra backprop, against FragSubset’s 2.29 steps without gradients. The COQR router is not reported on this subset (Sec. S1.7).

Paired bootstrap on this subset is in Sec. S2.8.

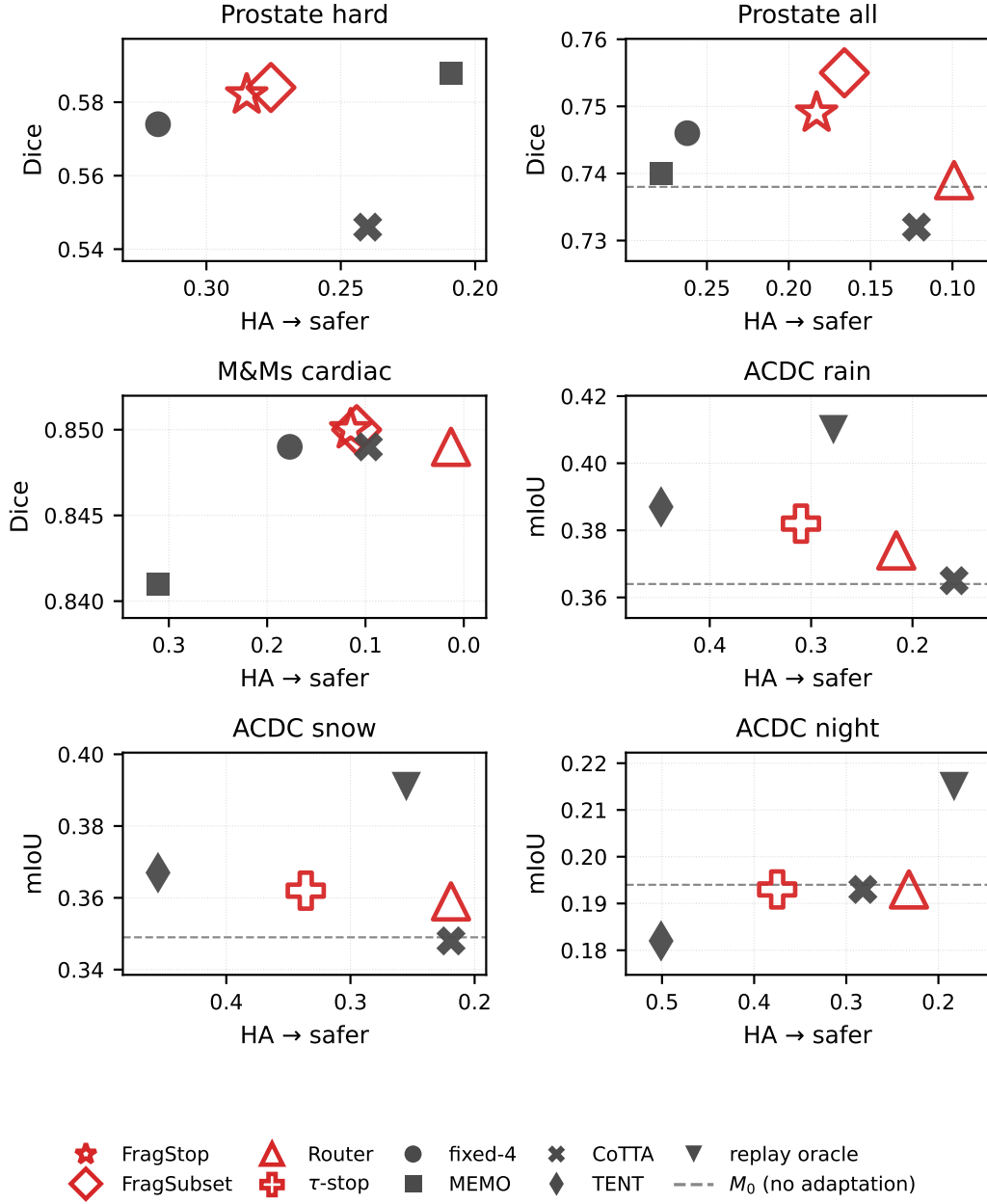


Figure S3: Reliability-accuracy trade-off (Tier C, descriptive) across six deployments. x : HA, axis inverted (right = safer); y : Dice or mIoU (never shared); colour marks ours (red) against baselines (grey) and shape identifies the method, as in the shared legend; marker size encodes nothing (ours are drawn larger for legibility), so deployed steps are read from the tables rather than from the figure; dashed = M_0 where a no-adaptation row is available (four of the six panels). The router is absent from prostate OOD-hard (an empty low bucket under the deployed cut-points); the replay oracle (per-case best-mIoU step, Sec. S2.8) appears only on the three ACDC panels where those trajectories are stored. Reading of the panels: Sec. S1.9.

S2.2 ACDC RAIN: STOP RULE ACROSS LEARNING RATES

The z-scored fragmentation stop cuts HA against the fixed-budget end along the entire budget axis; Table S6 gives the sweep. The effect is monotone in the learning rate: a larger budget produces more

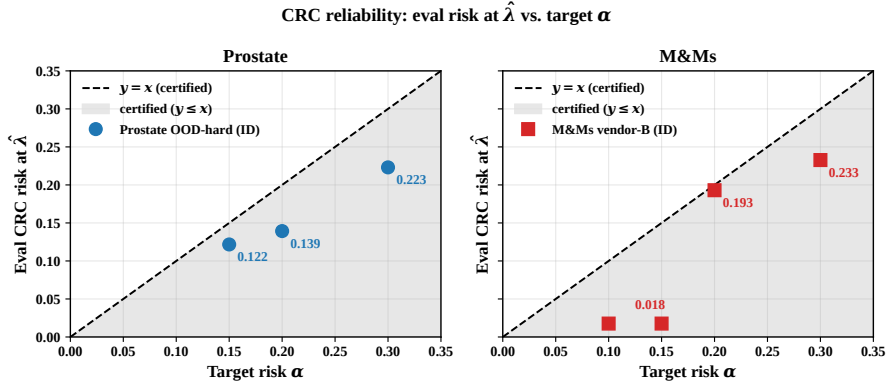


Figure S4: CRC reliability diagram. Target α vs. *eval* risk at $\hat{\lambda}$; the dashed diagonal is $y=\alpha$ and the shaded half below it is where the certificate holds ($y \leq \alpha$). Both benchmarks sit in the safe region at every α they are feasible at; prostate has no point at $\alpha=0.10$, where $n_{\text{cal}}=8$ puts the conformal floor above the budget (Sec. S2.5). Computed on the verification rerun described there.

Method	Dice	HA	Steps
M_0 (no adapt, ref.)	0.556	0	0
fixed-1	0.589	0.290	1.00
fixed-2	0.586	0.309	2.00
fixed-3 (<i>retrospective</i>)	0.581	0.257	3.00
fixed-4 (<i>default</i>)	0.574	0.318	4.00
FragStop (ours)	0.582	0.285	2.56
FragSubset (ours)	0.584	0.276	2.29
MEMO	0.588	0.209	4.00
CoTTA	0.546	0.240	4.00

Table S5: Prostate MRI OOD-hard ($n=34$), the hardest subset of OOD-all; online. The fixed- k rows are the trajectory checkpoints of Table 1; FragStop and FragSubset come from separate executions of the same protocol on the same cases, which reproduces those checkpoints at step 1 to 1×10^{-4} in Dice and 9×10^{-6} in δ , with identical M_0 predictions; region counts differ on two cases in the FragStop row (Sec. S9). **Bold** marks the best value per column among adaptive methods (the fixed- k ladder is reported for context).

over-adaptation, and the stop rule removes more of it. Each row is independently calibrated, with (μ_n, σ_n) and τ refit on the train split at that learning rate, so the trend is not one threshold reused across budgets. This is why we read the HA cut as tuning-robust rather than as an artifact of one learning-rate choice. The accuracy side moves little: τ -stop trails fixed-8 mIoU by 0.002 to 0.010 at every rate while cutting HA by 0.067 to 0.138.

lr	τ -stop (mIoU/HA)	fixed-8 (mIoU/HA)	HA cut	τ -stop steps
1×10^{-4}	0.368/0.152	0.370/0.219	−0.067	5.17
5×10^{-4}	0.375/0.255	0.385/0.336	−0.081	3.80
1×10^{-3}	0.378/0.272	0.388/0.393	−0.122	2.71
2×10^{-3}	0.382/0.310	0.387/0.448	−0.138	2.49

Table S6: ACDC rain: τ -stop vs. fixed-K across learning rates at a fixed $K=8$ (val, $n=100$); accuracy and deployed depth are reported beside HA, since HA alone is not interpretable (Sec. 3). Fixed-8 deploys 8 steps by definition. The 2×10^{-3} row is the main-text operating point.

The case bootstrap at the main operating point, covering both accuracy and reliability, is in Sec. S2.8.

S2.3 ABLATION CONFIGURATIONS AND THEIR NAMES

The budget-axis ablations are instances of one template: a modulation rule applied per step, optionally combined with the fragmentation stop. The names used in the main text are:

- **FragStop**: the stop alone, with no modulation. It halts at the first rise $\Delta n_k > 0$.
- **combo+param+Stop** (reported as **FragSubset**). The configuration key records the intended design: entropy weight down and Fisher weight up with r_k (**combo**), restriction of the updated parameter subset (**param**), and the stop. The configuration actually run, and reported in both main tables, is **mode=param_subset**: the entropy weight is held at 1.0 and the Fisher weight at $\lambda_{F,0}=1.0$, and the only risk-dependent quantity is the parameter subset (decoder normalization when $r_k > t_{\text{mid}}=0$, all normalization otherwise). The stored configuration also carries $\beta=0.7$ and $\lambda_{F,\text{max}}=4.0$; **neither is read in this mode**, and we list them only so the stored settings can be matched to this description. Both medical benchmarks use these settings (**fragopt_oodall**: Dice 0.7549, HA 0.1659, 2.677 steps, i.e. the **FragSubset** row of Table 4).
- **ent-weight+Stop**: partial modulation, with the entropy weight only, without the Fisher term or the parameter subset, plus the stop. The two continuous modulations referred to by the keys above scale the entropy weight down and the Fisher weight up with the risk, $w_{\text{ent}}(r_k)=\text{clip}(e^{-\beta r_k}, w_{\text{min}}, 1)$ and $\lambda_{\text{Fisher}}(r_k)=\text{clip}(\lambda_{F,0}e^{+\beta r_k}, \lambda_{F,0}, \lambda_{F,\text{max}})$; these are the forms β and $\lambda_{F,\text{max}}$ parametrize.
- **loss-switch+Stop**: the negative control, a discrete switch between the entropy and Fisher objectives instead of continuous modulation, plus the stop.

On prostate OOD-hard ($n=34$), against the fixed-4 row of Table S5 (HA 0.318): **combo+param+Stop** -0.043 , **FragStop** -0.033 , **ent-weight+Stop** -0.005 , **loss-switch+Stop** $+0.024$ (Dice 0.588). All four are differences against that single baseline, so each closes against the table. That baseline comes from a different execution (caption there); at 0.043 the largest is twenty times the ± 0.002 cross-execution scale of Sec. S9. The ordering is what the main text reads as the parameter-subset member carrying the reduction while modulating the entropy weight alone does not.

One further qualification is benchmark-specific. The parameter subset is the only risk-dependent quantity in this mode, and whether it varies at all depends on where r_k falls relative to $t_{\text{mid}}=0$. On prostate it varies: $r_k > 0$ on 23.7% of steps in the OOD-all run and on 11 of 78 (case, step) rows (14.1%) in the OOD-hard run behind Table S5. On M&Ms it does not: r_k ranges from -2.000 to -0.392 (median -1.869) and the condition fires on none of 788 rows, so both branches produce an identical adaptation configuration. The M&Ms **FragSubset** row of Table 3 is therefore not **FragStop** plus modulation; the two runs share their adaptation configuration and differ only in how the stop is evaluated: **FragStop** tests a single-step rise $n_k > n_{k-1}$, while this path scans the trajectory for the last step before the first rise. The residual HA difference (0.109 vs. 0.115, i.e. -0.0054) therefore has nothing to do with the modulation, and it comes from four cases out of 230: the two rules select different depths on two of them (-0.0037 of the total), and two further boundary regions flip between the runs, which are separate executions agreeing at step 1 to 1.6×10^{-4} in Dice (Sec. S9). The remaining 226 cases contribute $+0.0001$.

S2.4 A FIFTH EPISODIC OBJECTIVE: SAR

To test compilability beyond the TENT / EATA / CoTTA / MEMO trajectories, we wrap the controllers around SAR on prostate OOD-all under the identical protocol. We disable SAR’s EMA recovery: under our binary entropy scale its reset condition fires at every step, which collapses M_1 back to M_0 and makes the trajectory trivial. We did not instead rescale that threshold by $\ln C$, as Sec. S12 does for DeYO’s; the asymmetry is a limitation of this probe rather than a considered choice, and a rescaled reset may well keep the recovery mechanism alive.

SAR is a regime where stopping cannot create value, and we report it because that boundary is informative rather than because it is favorable:

- SAR falls below the un-adapted source from the first step (Dice 0.734 at $k=1$ against M_0 0.738), so there is no early segment worth keeping.

- It accumulates the highest HA of any objective we evaluate on prostate OOD-all (0.374 at $k=4$; the driving benchmark reaches higher still, Table S7).
- FragStop tracks the ladder rather than beating it: 0.724/0.339 at 2.06 steps, worse than SAR fixed-1 (0.734/0.300) on both axes; Δ HA vs. fixed-4 is -0.035 $[-0.078, +0.009]$, crossing zero.
- The oracle stop on the same trajectories reaches 0.753/0.266, so per-case structure does exist here that fragmentation does not recover.

This bounds the stop rule’s applicability: it requires a trajectory with an early segment worth keeping. The router is unaffected by this bound, since rollback does not depend on an improving prefix.

S2.5 CERTIFIED OPERATING POINT: A MONOTONE SURROGATE

The surrogate that is certified. Conformal risk control (CRC) (Vovk et al., 2005; Angelopoulos et al., 2024) requires a risk *monotone* in the gate threshold, and deployment HA is not: the dynamic-denominator property that makes it non-monotone in the budget (Sec. 3.1) acts here in the gate variable, since $|D(\lambda)|$ shrinks as the gate rejects edits, so rejecting more can leave HA flat or raise it. Fixing the denominator to $|D_{\max}|$ and scoring each region by its mean entropy drop gives a strictly monotone risk to which CRC applies unchanged. The obstruction is specific to monotone CRC: distribution-free procedures for non-monotone bounded losses, such as Learn-Then-Test, could in principle certify deployment HA itself on a discretised threshold grid, and we did not run one. The certificate therefore covers the fixed-denominator surrogate, *not* the deployed dynamic HA.

The main text states the correctness result (once the fixed-denominator surrogate replaces deployment HA, the region gate certifies at the requested budget) and defers the operating points here ($\hat{\lambda}$ is calibrated as in Sec. S1.7).

What this section is computed on. The certificate needs per-voxel probabilities, which the archived trajectories do not store, so the operating points below come from a second verification rerun of the same EATA trajectories as the main tables, separate from the one in Sec. S9 and checked against the archived runs by the same criteria: on cardiac, step-1 gradient norms match to 2.7×10^{-7} and cohort Dice and HA to 2.4×10^{-6} and 1.5×10^{-4} . Two conventions differ from the rest of the paper and are worth stating. First, only cases whose step-1 edit contains a component above the minimum-size threshold can be gated at all, since the gate has nothing to accept or reject otherwise: 168 of the 230 cardiac evaluation cases and 10 of the 20 calibration cases qualify, against 34 of 34 and 8 of 8 on prostate. Second, the certified regions are connected components of the raw multi-class disagreement, whereas HA on the medical benchmarks scores components of the foreground-binarized disagreement, so a class-to-class edit is a region here and is not one there; on cardiac the two counts are 168 and 167 on the same masks. The 167 here is computed on the verification rerun (Sec. S9); the archived count at $k=1$ in Table S15 coincides, though the two runs differ by one case at $k=2$ and $k=3$ (Sec. S8).

- *Prostate OOD-hard.* Certifies at all three feasible budgets, with eval risk 0.122, 0.139 and 0.223 against calibration bounds of 0.142, 0.184 and 0.270; $\alpha=0.10$ is infeasible at $n_{\text{cal}}=8$ (below).
- *M&Ms vendor-B.* Certifies at all four, with eval risk 0.018 at $\alpha=0.10$ and 0.15, 0.193 at 0.20 and 0.233 at 0.30.
- *Reliability diagram.* Fig. S4 plots target α against eval risk at $\hat{\lambda}$. Both benchmarks certify at every α they are feasible at, and feasibility is what separates them: the two sit on opposite sides of the same $1/(n_{\text{cal}}+1)$ floor (below).

Two boundaries are inherent to CRC at small holdouts, and neither is specific to fragmentation:

- *Feasibility is bounded by n_{cal} .* The prostate certificate is calibrated on the 8 OOD-hard cases held out of the 34-case evaluation split (the remainder of the 42-case pool, Sec. S1.9) and evaluated on the 34. At $\alpha=0.10$ with $n_{\text{cal}}=8$ the conformal bound is at best $1/(n_{\text{cal}}+1)=1/9=0.111 > \alpha$ even if every region is rejected, so no gate can certify. The M&Ms certificate is calibrated on the 10 scorable cases of the 20-case vendor-B calibration

split, where the same floor is $1/11=0.091$ and $\alpha=0.10$ clears it by 0.009. The two benchmarks therefore fall on opposite sides of one arithmetic bound, which is what makes this a property of the calibration set size rather than of either dataset. It relaxes on the larger ACDC splits.

- *The certified surrogate is deliberately not deployment HA.* At the same $\hat{\lambda}$, gated deployment HA runs two to four times the certified surrogate risk (prostate $\alpha=0.20$: 0.367 against 0.139, $2.6\times$; M&Ms $\alpha=0.10$: 0.070 against 0.018, $4.0\times$). Across the seven feasible operating points (prostate $\alpha=0.10$ is infeasible at $n_{\text{cal}}=8$) the ratio ranges from $1.7\times$ to $4.0\times$, the widest gap being M&Ms at $\alpha=0.10$ and 0.15. We report the deployment number openly so the certificate is read for what it guarantees, a monotone region-risk bound on the fixed-denominator surrogate, rather than mistaken for a direct bound on deployment HA. The gap is the intended output of the dual-metric design.

S2.6 PROSTATE QUALITATIVE ROW

The main text carries the M&Ms row of the “harm invisible to Dice” figure; Fig. S5 is its prostate companion, selected by the protocol of Sec. S1.15.

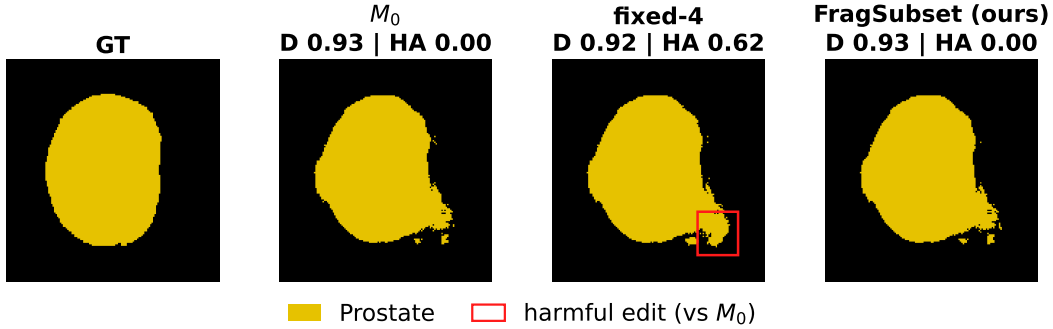


Figure S5: Harm invisible to Dice, prostate row (FragSubset, prostate_B_0017); the companion of the main-text M&Ms row. L→R: GT (evaluation reference only); M_0 ; fixed-4, where the red box locates the harmful accepted edit it introduces relative to M_0 (the flagged component is 130 pixels on this slice, under 1% of the panel, so we mark its position rather than trace a boundary too small to read); and FragSubset. Panel D/HA are *slice-level*, not case means: slice M_0 D=0.93; fixed-4 D=0.92 at HA=0.62; FragSubset keeps HA=0. Selection protocol in Sec. S1.15.

S2.7 ACDC: ALL FOUR CONDITIONS

The main text discusses night (the hard condition, where M_0 is near-collapsed) and gives the four-condition HA cascade inline; Table S7 is the full panel, and rain is the condition where adaptation helps most. Fog and snow behave like rain: the cascade $\text{TENT}_8 \rightarrow \tau\text{-stop} \rightarrow \text{Router}$ lowers HA at a small mIoU cost, and CoTTA holds HA down only by sitting at M_0 , which it does on all four conditions. On rain the HA reduction grows with the budget ($\Delta\text{HA} -0.138$ on rain at $\text{lr } 2 \times 10^{-3}$, against -0.067 at $\text{lr } 10^{-4}$).

On fog the source model is strongest and the signal correspondingly quiet. Under the mild budget ($\text{lr } 10^{-4}$, $K=4$) the stop rule runs the full budget on 49% of cases, and $\tau\text{-stop}$ mIoU (0.4751) sits within 0.002 of both fixed-K (0.4755) and the validation oracle (0.4766): fragmentation does not over-fire when there is nothing to stop. The hard tertile is again where M_0 is weakest (fog 0.456 against 0.477 in the low tertile).

S2.8 REMAINING PAIRWISE INTERVALS

This section is the single authoritative list of *deployment-level controller-vs-budget* paired intervals (excluding the SAR probe of Sec. S2.4, the frozen-design test of Sec. S10, and the backbone and objective extensions of Secs. S11 and S12, each reported in place): every such comparison quoted

Split	M_0	TENT ₈	CoTTA	τ -stop	Router
fog	0.474	0.486/0.445	0.473/0.171	0.483/0.211	0.481/0.147
rain	0.364	0.387/0.448	0.365/0.159	0.382/0.310	0.374/0.216
snow	0.349	0.367/0.455	0.348/0.219	0.362/0.336	0.359/0.219
night	0.194	0.182/0.501	0.193/0.282	0.193/0.375	0.193/0.232

Table S7: ACDC, all four conditions (mIoU/HA; lr 2×10^{-3} , $K=8$; $n=100$ for fog/rain/snow, $n=106$ for night). In the roles easy / main / cross / hard, fog and snow are the easy and cross conditions; rain and night are the two the main text discusses. TENT₈ is TENT run to the fixed budget with no control; M_0 takes no step, so its HA is 0 by construction. Router steps per case: 0.96 (fog), 0.97 (rain), 1.01 (snow), 1.67 (night).

anywhere in the paper appears here once, so a number has exactly one source. Matched-control intervals (Sec. S3) and calibration-sensitivity bands (Sec. S7) are different comparisons and live in their own sections. All are $B=5000$, seed 1000, paired on cases. Point estimates are computed on unrounded case-level values, so each may differ from the corresponding three-decimal difference of the main tables by up to 0.001; a larger gap would indicate that the per-case source and the table are not the same file.

M&Ms vendor-B ($n=230$).

- Router vs. deployable fixed-2: $\Delta\text{HA} -0.151$ $[-0.191, -0.115]$ (Tier-A), at $\Delta\text{Dice} -0.0015$ $[-0.0032, +0.0001]$.
- Router vs. retrospective fixed-3: $\Delta\text{HA} -0.116$ $[-0.152, -0.081]$, Wilcoxon $p=9.1 \times 10^{-10}$, at $\Delta\text{Dice} -0.0008$ $[-0.0022, +0.0006]$; about $5.8 \times$ the corresponding FragSubset-vs-fixed-3 margin on this split.
- Router vs. default fixed-4: $\Delta\text{HA} -0.163$ $[-0.202, -0.126]$ (Tier-A'), at $\Delta\text{Dice} -0.0000$ $[-0.0014, +0.0012]$.
- Router vs. CoTTA: $\Delta\text{HA} -0.084$ $[-0.113, -0.055]$ at $\Delta\text{Dice} +0.0003$ $[-0.0004, +0.0009]$.
- Router vs. FragSubset: $\Delta\text{HA} -0.096$ $[-0.130, -0.062]$, at a third of the deployed updates, or about half the executed ones.
- FragSubset vs. deployable fixed-2: $\Delta\text{HA} -0.055$ $[-0.084, -0.028]$ (Tier-A).
- FragSubset vs. retrospective fixed-3: $\Delta\text{HA} -0.020$ $[-0.040, -0.000]$ (the upper bound is below zero but rounds to it, which is why $P=0.978$ rather than 1.0; barely separated, Tier-B).

Prostate OOD-all ($n=93$).

- Router vs. default fixed-4: $\Delta\text{HA} -0.162$ $[-0.227, -0.098]$ (Tier-A'), 61/93 wins, at $\Delta\text{Dice} -0.0077$ $[-0.0236, +0.0072]$.
- Router vs. deployable fixed-1: $\Delta\text{HA} -0.142$ $[-0.194, -0.094]$, $P(\Delta\text{HA} < 0)=1.0$, at $\Delta\text{Dice} -0.0130$ $[-0.0252, -0.0022]$ (the accuracy interval excludes zero, unlike every Dice comparison on M&Ms).
- Router vs. retrospective fixed-3: $\Delta\text{HA} -0.072$ $[-0.139, -0.007]$, $P(\Delta\text{HA} < 0)=0.99$, 42/93 wins, at $\Delta\text{Dice} -0.0122$ $[-0.0279, +0.0020]$.
- FragSubset vs. default fixed-4: $\Delta\text{HA} -0.096$ $[-0.143, -0.049]$ (Tier-A').
- FragSubset vs. deployable fixed-1: $\Delta\text{HA} -0.076$ $[-0.113, -0.041]$ (Tier-A; $P=1.0$, Wilcoxon $p=5.7 \times 10^{-5}$, 48/93 wins).
- FragSubset vs. retrospective fixed-3: $\Delta\text{HA} -0.006$ $[-0.046, +0.035]$ (CI crosses zero; not claimed), at $\Delta\text{Dice} +0.004$ $[-0.003, +0.012]$.
- FragStop vs. retrospective fixed-3: $\Delta\text{HA} +0.011$ $[-0.023, +0.047]$ (directionally worse; not claimed).

Prostate OOD-hard ($n=34$). Panel in Sec. S2.1.

- FragSubset vs. default fixed-4: $\Delta\text{HA} -0.043 [-0.086, -0.001]$ ($P(\Delta\text{HA}<0)=0.98$; marginal at the upper bound).
- FragSubset vs. retrospective fixed-3: $\Delta\text{HA} +0.019 [-0.025, +0.068]$ (point estimate reverses; not claimed).
- FragSubset vs. FragStop (modulation beyond stopping): Wilcoxon $p=0.016$ but Holm-adjusted $p=0.065$ with a bootstrap CI crossing zero. We label this *not established*; it is the only ordering claim we make between two of our own controllers, and the only place Holm correction is applied (Sec. S1.9).

ACDC rain ($n=100$, $lr\ 2\times 10^{-3}$, $K=8$).

- Router vs. the HA-best fixed-1: $\Delta\text{HA} -0.048 [-0.088, -0.012]$ (Tier-A); vs. fixed-8: $\Delta\text{HA} -0.233 [-0.281, -0.187]$.
- τ -stop vs. fixed-1: $\Delta\text{HA} +0.047 [+0.025, +0.072]$ (trades HA for steps/mIoU against the HA-optimal fixed budget); vs. fixed-8: $\Delta\text{HA} -0.138 [-0.172, -0.105]$ ($P=1.0$; the main operating point of the learning-rate sweep, Sec. S2.2).
- τ -stop vs. fixed-8, accuracy side: $\Delta\text{mIoU} -0.0053 [-0.0124, +0.0017]$ ($\text{CI} \ni 0$), so the HA cut is mIoU-neutral.
- ACDC snow, same setting: τ -stop vs. fixed-8 $\Delta\text{HA} -0.119 [-0.155, -0.085]$ ($P=1.0$) at $\Delta\text{mIoU} -0.0053 [-0.0127, +0.0018]$.
- Router vs. τ -stop on rain, accuracy side: $\Delta\text{mIoU} -0.008 [-0.014, -0.003]$, the one condition where routing on top of stopping costs significant accuracy.
- ACDC night ($n=106$): Router vs. TENT_8 $\Delta\text{mIoU} +0.011 [+0.004, +0.018]$ at $\Delta\text{HA} -0.269$; both are damage control against a TENT run that falls below M_0 , not an adaptation gain.

ACDC replay oracle. A replay oracle that selects, per case, the best-mIoU point along the stored aggressive-budget trajectory reaches mIoU 0.410 at HA 0.278 on rain, 0.391 at 0.255 on snow and 0.215 at 0.183 on night, at 3.5, 3.5 and 2.4 deployed steps respectively. It is an upper bound on *accuracy*, not on *reliability*: on rain the router sits below it on HA (0.216 against 0.278) while the τ -stop does not (0.310). We read oracle numbers as bounds rather than as stable per-case labels, because on plateau-like trajectories the utility surface over $\{M_k\}$ is nearly flat, so the arg-max step need not be unique; we never train against them.

S3 MATCHED-CONTROL DECOMPOSITION ACROSS REGIMES

The main-text tables compare controllers against *budgets*. This section asks a different question, the one a reader who suspects the metric can be gamed will ask: at the router’s own rollback rate, how much of the HA reduction is attributable to the *fragmentation score* rather than to the act of declining to adapt? Every alternative selector below deploys exactly the same actions at exactly the same bucket sizes; only the score that assigns cases to buckets changes. All replays reproduce the deployed controller’s own row to the precision at which the main tables print it, which is the calibration that licenses the counterfactuals: prostate 0.7387/0.0994/0.548 $\rightarrow$ 0.739/0.099/0.55 with buckets 27/24/42, and M&Ms 0.8489/0.0133/1.104 $\rightarrow$ 0.849/0.013/1.10 with buckets 130/46/54.

M&Ms: fragmentation separates from every alternative. No random assignment out of 1,000 reaches the router’s HA. Paired bootstrap over cases ($B=5000$) against each informed alternative: $\Delta\text{HA} -0.040 [-0.065, -0.018]$ vs. entropy, $-0.043 [-0.070, -0.019]$ vs. MSP, and $-0.032 [-0.057, -0.009]$ vs. augmentation agreement, at ΔDice of -0.0007 , -0.0005 and -0.0001 respectively. The margin over the strongest alternative is $16\times$ the run-to-run reproducibility of a cohort HA mean (Sec. S9), and the consistency baseline pays eight extra forward passes per step for a signal that fragmentation obtains from the two masks it already has.

Prostate: informative, but not distinguishable from confidence. At the router’s rollback rate (42/93), fragmentation-scored routing gives HA 0.0994 against 0.1325 [0.1053, 0.1607] for random assignment, 0.1044 for entropy and 0.1030 for MSP. Fragmentation beats random ($P=0.010$ over

Selector (same actions, buckets 130/46/54)	Dice	HA	Steps
Fragmentation (δ , deployed)	0.8489	0.0133	1.104
Augmentation agreement (8 forward passes)	0.8490	0.0451	1.104
Entropy	0.8496	0.0537	1.104
MSP	0.8494	0.0567	1.104
Random (mean of 1,000 draws)	0.8494	0.0605	1.104
Oracle on true HA at step 1	0.8503	0.0023	1.104
Oracle on true Δ Dice	0.8515	0.0217	1.104
M_0 (no adaptation)	0.8489	0	0

Table S8: Matched-bucket controls on M&Ms vendor-B ($n=230$). Every selector row rolls back the same number of cases and deploys the same per-bucket depths; only the ranking score differs.

1,000 draws), so the score carries real information here too, but paired bootstrap against the confidence proxies gives Δ HA -0.0050 $[-0.0572, +0.0475]$ and -0.0036 $[-0.0569, +0.0497]$, intervals that straddle zero, while Δ Dice is -0.0110 $[-0.0206, -0.0026]$, which does not. The two hard buckets overlap on only 24 of 42 cases, so this is not one partition perturbed. The reading consistent with Sec. 5.2 is that the HA cut on this split comes mostly from rolling back difficult cases, and several risk-correlated signals rank difficulty.

ACDC: a controlled substitute, and a null result. The archived per-case records for ACDC store only the action each case actually received, so a rolled-back case has no recorded “had it run” outcome and the full counterfactual is not available offline. We therefore report a narrower experiment: rollback maps to M_0 and every retained case to the τ -stop outcome, which *is* recorded for all cases. **This does not reproduce the deployed ACDC router** (whose retain action is the parameter-free Δn_k stop rule within a per-bucket cap, not the global τ -stop; the per-condition tertiles assign buckets, they are not the stop rule), and its numbers are not comparable with the main-text ACDC table; it isolates one question only: given a fixed rollback quota and a fixed retain action, does δ pick the right cases? Selecting the top- n_{hard} cases by δ reproduces the deployed hard bucket exactly in all four conditions (18/21/29/43), so the selector is faithful.

Condition	n	n_{hard}	δ HA	Random HA [95%]	Oracle HA	$P(\text{rand} \leq \delta)$
fog	100	18	0.150	0.173 [0.156, 0.188]	0.106	0.005
rain	100	21	0.242	0.245 [0.221, 0.266]	0.147	0.366
snow	100	29	0.256	0.238 [0.214, 0.262]	0.136	0.934
night	106	43	0.206	0.222 [0.198, 0.246]	0.113	0.087
pooled	—	—	0.213	0.220	0.126	—

Table S9: Rollback-selection control on ACDC at the aggressive budget (retain action fixed to τ -stop). δ -based selection is not better than random overall, fog excepted.

The null is not a surprise but a prediction: at this budget δ ’s association with HA is weak (0.139) and n_{reg} ’s has inverted (Sec. S4), and a score that barely correlates with harm cannot rank cases by it. The routing increment the main text reports at this budget therefore comes chiefly from the rollback quota rather than from case selection. A full counterfactual would require re-running the ACDC trajectories under each alternative assignment; we list this as a contingency rather than a claim.

Prostate158: the quota, not the ranking. The frozen-design test of Sec. S10 is the one deployment in this paper whose benchmark took no part in the design’s selection, so it is also the cleanest place to ask what the score contributes there. We replay it under the protocol used above: bucket sizes 48/39/39 and depths 3/2/0 held fixed, only the ranking score changed. The replay reproduces the deployed arm: cut-points refit from the 32-case calibration split agree with the archived values to the last digit, assignment by cut-point and by rank agree on all 126 evaluation cases, and the four fixed- k rows reproduce Table S20 to printed precision.

The deployed coordinate is not better than random here. Fragmentation (δ at step 1) reaches HA 0.161 against 0.164 [0.144, 0.184] for random assignment at the same quota, and 39.5% of 1,000

random draws match or beat it; entropy (0.161), n_{reg} (0.157) and gradient norm (0.170) all fall inside that band, and none separates on Dice. What this isolates is the ranking alone: no shrink outcome is recorded for cases the deployed design did not send to the mid bucket, so every arm is compared at depths 3/2/0, and the shrink action contributes a further -0.022 HA independently of how cases are ordered. The frozen design’s advantage over that benchmark’s own retrospective-best budget is therefore unaffected; what does not survive is the attribution of that advantage to case selection.

As on ACDC, the null is a prediction rather than a surprise, and here it is a quantitative one. Measured on the same trajectories, with the pipeline that reproduces the published pooled prostate entries ($n=124$) of Table 2 to the last digit, δ ’s level correlation with HA is 0.156 here, against 0.557 on that pooled prostate set and 0.516 on cardiac, while its step-differenced correlation is 0.449, in line with those benchmarks (0.515 and 0.347). The router ranks cases by the level, and the level is what has collapsed; n_{reg} is weaker still (0.050), so its narrow edge above is noise and not a coordinate preference. The case-level ranker of Table S14 agrees: scored the same way, the step-1 signal ranks top-tertile HA at 0.475 (δ) and 0.506 (n_{reg}) here, against 0.686 and 0.685 (δ , n_{reg}) on the pooled prostate set and 0.692 and 0.716 on cardiac. What the budget axis uses, the step-to-step rise, is intact, which is consistent with the frozen design still clearing the fixed budget.

Pre-specification reconciliation. Criteria for the M&Ms and prostate replays were written to `operation.md` before those numbers were computed.

Pre-specified expectation	M&Ms	Prostate
Random at matched rate is worse	held (0/1,000)	held ($P=0.010$)
Entropy/MSP intermediate or not better	held, and significantly	held in point estimates; difference not significant
Oracle is an upper bound	failed (see below)	held

Table S10: Pre-specification reconciliation for the M&Ms and prostate matched-control replays.

The third expectation specified an oracle that buckets by true ΔDice . That oracle reaches HA 0.0217 on M&Ms, *worse* than the deployed fragmentation router’s 0.0133, so it is not an upper bound on HA. The specification, not the execution, was wrong: an oracle bounds only the quantity it optimizes, and a ΔDice oracle optimizes Dice-level harm. The HA oracle (0.0023) is the correct bound. We report the failure rather than substituting the working oracle silently, and note that it is itself an instance of this paper’s thesis: selecting cases with perfect knowledge of the Dice outcome still accepts nearly ten times the harmful area that selecting on HA would.

Three independent lines, one regime split. The decomposition above is a deployment-level replay. Two further analyses with no methodology in common reach the same partition.

Analysis	M&Ms	Prostate	Prostate158	ACDC (aggr.)
Deployment replay (this section)	separates from every alternative	not distinguishable from confidence	not better than random	not better than random
Case-level ranking AUROC (Sec. S4)	0.716 vs. grad-norm 0.616	0.685 $\approx$ 0.681 $\approx$ 0.684	0.475 and 0.506, — at chance	—
Calibration perturbation (Sec. S7)	beats fixed-3 in 100% of resamples at $n_{\text{cal}}=5$	$\sim 1/5$ – $1/3$ of re-samples lose to fixed-3	not run	router’s value at or above the band center

Table S11: The same three-regime split recovered by deployment replay, case-level ranking and calibration-perturbation propagation. Prostate158 falls in the same regime as ACDC on the deployment replay, the only analysis run on both; the calibration-perturbation propagation was not run on it.

S4 CLUSTERED INTERVALS, PER-STEP STRUCTURE, AND CASE-LEVEL RANKING

The pooled correlations in the main text treat the four (case, step) rows of one case as independent observations. They are not: the steps of a case share an image, a source prediction and one adaptation

trajectory. Every interval below therefore resamples *cases*, carrying all of a case’s steps together ($B=5000$).

Benchmark	Signal	ρ vs. HA [95%]	$\rho(\Delta, \Delta\text{HA})$ [95%]
Prostate, $n=124$	n_{reg}	0.594 [0.481, 0.686]	0.464 [0.352, 0.563]
	δ	0.557 [0.428, 0.659]	0.515 [0.416, 0.606]
	entropy	0.119 [-0.021, 0.248]	-0.290 [-0.399, -0.181]
	MSP	-0.161 [-0.296, -0.010]	0.278 [0.170, 0.385]
	grad-norm	0.270 [0.175, 0.354]	-0.035 [-0.108, 0.034]
	GraTa-cos	-0.093 [-0.208, 0.027]	-0.009 [-0.106, 0.086]
Cardiac, $n=230$	n_{reg}	0.597 [0.532, 0.657]	0.455 [0.377, 0.527]
	δ	0.516 [0.439, 0.585]	0.347 [0.266, 0.421]
	grad-norm	0.146 [0.082, 0.206]	0.035 [-0.031, 0.101]
	entropy	0.032 [-0.069, 0.130]	-0.006 [-0.082, 0.073]
	MSP	-0.012 [-0.108, 0.087]	0.017 [-0.060, 0.092]
	1-agreement (aug.)	0.264 [0.161, 0.362]	0.051 [-0.037, 0.138]
	1-agreement (dropout)	0.047 [-0.064, 0.157]	0.016 [-0.073, 0.105]
Driving mild, $n=406$	n_{reg}	0.503 [0.445, 0.557]	0.290 [0.236, 0.341]
	δ	0.478 [0.419, 0.534]	0.238 [0.168, 0.306]
Driving aggr., $n=306$	n_{reg}	-0.147 [-0.238, -0.053]	0.046 [0.001, 0.091]
	δ	0.139 [0.051, 0.230]	0.187 [0.129, 0.243]

Table S12: Case-clustered bootstrap intervals for every signal–HA correlation reported in the main text. Confidence proxies contain zero on cardiac, and on prostate only entropy’s level interval does; so does the injected-dropout agreement; the sign inversion at the aggressive driving budget does not.

Per-step structure, and what pooling costs. Pooling can misrepresent a signal whose scale drifts across steps, and one of ours is affected.

Benchmark	Signal	$k=1$	$k=2$	$k=3$	$k=4$	pooled
Prostate	n_{reg}	0.522	0.679	0.653	0.428	0.594
	grad-norm	0.460	0.586	0.551	0.144	0.270
Cardiac	n_{reg}	0.527	0.560	0.675	0.603	0.597
	grad-norm	0.332	0.164	0.384	0.190	0.146
Driving mild	n_{reg}	0.499	0.428	0.388	0.405	0.503

Table S13: $\rho(\text{signal}, \text{HA})$ computed within each step. Fragmentation is positive at every step of every benchmark, so its pooled value is not an artifact of pooling. Grad-norm on prostate is comparable to fragmentation through $k=3$ and collapses at the horizon; pooling compresses that into a single low number, which understates it. We report the pooled protocol in the main table for comparability and this decomposition alongside it.

At the aggressive driving budget the inversion is present from the first step ($-0.101, -0.218, -0.218, -0.171, -0.142, -0.153, -0.147, -0.141$ for $k=1 \dots 8$): the signal is not degrading along the trajectory, it is wrong at that learning rate from the outset. Restricting those same trajectories to $k \leq 4$, matching the depth of the mild-budget runs, does not restore it (-0.054 rain, -0.222 snow, -0.142 night), which is what attributes the collapse to update size rather than trajectory length. Per condition at the mild budget, $\rho(n_{\text{reg}}, \text{HA})$ is 0.462 (fog), 0.420 (rain), 0.321 (snow) and 0.372 (night), with Δ columns 0.275, 0.256, 0.269 and 0.246; the corresponding δ level values are 0.393, 0.361, 0.334 and 0.359. Per-step signal statistics at the aggressive budget were not stored for fog, which is why the pooled aggressive row covers three conditions.

Case-level ranking: which harm is predictable. A correlation over (case, step) pairs does not say whether a step-1 reading ranks *cases* by the harm that deployment will incur. We therefore score each case by its step-1 signal and ask how well that ranks two different notions of harm at the deployed budget.

Benchmark	Signal	AUROC, $\Delta\text{Dice}<0$	AUROC, HA in top tertile
Prostate	n_{reg}	0.464 [0.36, 0.56]	0.685 [0.59, 0.77]
	δ	0.446	0.686
	entropy	0.555	0.681
	grad-norm	0.441	0.684
	MSP	0.435	0.297 (inverted)
Cardiac	n_{reg}	0.434	0.716 [0.65, 0.78]
	δ	0.425	0.692
	grad-norm	0.374 [0.30, 0.45]	0.616 [0.54, 0.69]

Table S14: Step-1 signal as a case-level ranker, scored against two notions of deployment harm. Orientation is fixed to “higher signal $\Rightarrow$ positive”; MSP therefore ranks in the inverted direction, which is itself informative. No signal ranks Dice degradation in the stated direction, cardiac grad-norm ranking it inverted; every signal ranks HA-level harm.

First, the two harms are different quantities and only one of them is predictable, which is the case-level counterpart of the measurement-level claim in Sec. 3. The sharpest single instance is grad-norm on cardiac: AUROC 0.374 with an interval excluding 0.5, i.e. a *high* gradient norm predicts a Dice *improvement*, on cases whose local edits are meanwhile harmful. Second, the ranking view recovers the regime split of Sec. S3 without sharing any machinery with it. On prostate the four informative signals are indistinguishable at the 0.68 level, on cardiac fragmentation separates from grad-norm.

S5 WHICH HALF THE ROUTER DECLINES, AND WHAT IT COSTS

Each signal deployed as a stop rule. Table 2 compares the signals as *predictors*; deploying each as a stop rule on prostate OOD-hard ($n=34$) compares them as *controllers*. Fragmentation stops at HA 0.285 at Dice 0.582 with no backward pass; grad-norm reaches 0.295 at Dice 0.593, so the prediction-space signal matches it on harm while giving up a little overlap, which is the trade it makes against a gradient-space one. The entropy and MSP variants reach 0.293 and 0.291. All four therefore land within 0.010 HA of each other. These are unpaired point estimates on 34 cases and we do not read the ordering among them as significant; what the comparison establishes is that a read-only signal is not worse as a stop rule, not that it is better.

Which half it declines. Rolling back to M_0 gives HA=0 trivially but throws away accuracy (0.738 vs. FragSubset’s 0.755 on OOD-all), so the goal is *Dice-matched* HA reduction. The router rolls back 130 of 230 M&Ms cases and 42 of 93 prostate cases, and we read that as the intended behavior rather than as a defect: *whether this case should be adapted at all* is the question the paper poses, and a controller that answers “no” for half a cohort is answering it. What separates this from the degenerate solution is which half: on M&Ms the retained cases are net-helped and Dice stays at M_0 while HA falls to 0.013, whereas on prostate they are not (main text, prostate section). Indiscriminate abstention would sit at M_0 on both axes and order nothing. It is worth stating the comparison against M_0 plainly. On cardiac nobody wins it: fixed-4 adaptation is worth +0.00006 Dice in aggregate and the router +0.000016, both indistinguishable from doing nothing. On prostate fixed-1 does win it (+0.014), and the router, at +0.0008, declines nearly all of that gain. That is the claim the main text makes at this point (Sec. 5.1). Where a stream does have something to gain the router keeps it: on ACDC it adds +0.007 to +0.010 mIoU over M_0 on fog, rain and snow, and gives up 0.0015 on night (−0.002 at the precision of Table 5), the condition where M_0 is near-collapsed. Nor is the reduction available to any equally conservative rule: at the router’s own rollback rate on M&Ms, random assignment yields HA 0.060 [0.043, 0.080] (all 1,000 draws above the router’s 0.013), confidence-scored routing 0.054–0.057, and augmentation-agreement routing 0.045 at eight forward passes per step; fragmentation reaches 0.013 with none, at ΔDice within 0.001 of every alternative (Sec. S3). The deployed edit set is small (coverage $\delta=3\times 10^{-5}$; 17 of 230 cases retain supra-threshold regions, with within-region HA 0.180 and BA 0.248; Sec. S6): the router’s 0.013 reflects deploying few edits and choosing them well, not many edits that happen to be harmless.

The signal is not the metric. The controllers do not read or optimize HA at decision time: HA requires ground truth (Eq. 1), whereas they act on label-free fragmentation, and the two correlate at ρ

between 0.478 and 0.597 across benchmarks and coordinates at comparable budgets (Table 2), not 1.0. The negative control of Sec. S2.3, which *raises* HA, confirms that the reduction follows from reading the right signal.

Cost. Fragmentation costs ~ 37 ms/step with no backprop, against ~ 150 ms and $+2.5$ GB for gradient-norm. Our controllers cut HA against the fixed budget without extra backprop, and the router does so at 0.55–1.10 deployed updates per case (1.00–1.67 SGD steps once the discarded probe is counted, against 4.00). CoTTA also cuts HA without extra backprop, at M_0 ’s Dice on M&Ms and below it on prostate. The certified operating point for the monotone surrogate of Sec. 4.2 is in Sec. S2.5.

S6 EDIT-AREA ACCOUNTING: HARMFUL, BENEFICIAL AND NEUTRAL

HA reports the harmful share of the edited area. The same partition has two other cells, and reporting all three changes what the number means. For each disagreement region we compare region-level error under M_k and under M_0 : the region is harmful if error rises, *beneficial* (BA) if it falls, and neutral if it is unchanged. All three are fractions of the same denominator $|D|$, so $HA + BA \leq 1$ and the remainder is neutral area. “Neutral” covers two different things and the table does not separate them: disagreement voxels that lie in a region large enough to be scored but whose region-level error is unchanged, and disagreement voxels that lie in no scored region at all because every component containing them falls below the 16-voxel minimum. The second is not a small residual: in 376 of the 920 cardiac (case, step) rows the disagreement set is non-empty but no component survives the threshold, so the whole edit is unscored. A reader should therefore read neutral area as “not judged harmful” rather than as “judged harmless”. This is a decomposition of *area*; Table 5 decomposes the same cases by *case-level* Dice change. The two are not interchangeable, and Table S14 is the reason: the harm visible in one is not the harm predictable from the signal in the other.

Because $|D|$ moves with the gate (the dynamic-denominator property of Sec. 3), $BA - HA$ differences are not comparable between controllers that deploy different amounts of edit. The comparable quantity is the ratio BA/HA , and we report coverage δ so the size of the denominator is visible. We also report each row twice: over all cases, and over the cases that actually contain a region above the minimum-size threshold.

Three things the table makes visible that HA alone does not. *Adaptation repairs more than it breaks in the early steps and reverses at the horizon*: on cardiac the fixed budgets run BA/HA 1.26, 1.37, 1.23 and then 0.84 at $k=4$, which is the accounting form of the non-monotonicity of Sec. 3.1. Mean overlap is flat across that reversal. *The router’s low HA is mostly abstention*: only 17 of 230 cardiac cases retain a supra-threshold region at all, and coverage falls by an order of magnitude, so 0.013 reports deploying few edits with a favorable composition rather than many harmless ones. *The low-fragmentation bucket is the net-harmful one* on both medical benchmarks (BA/HA 0.00 on cardiac and 0.31 on prostate) while the mid bucket is net-beneficial (2.44 and 1.35; on cardiac the low bucket is also the one allowed to run longest, at depth 3, whereas prostate deploys both at depth 1): the score orders where to do less, not where more would help, which is the quantitative form of the limitation stated in Sec. 5.2.

Accuracy, at full precision. Table 3 prints the router and M_0 at the same three-decimal Dice on cardiac, which invites the question of whether that is a rounding coincidence. It is not: the paired difference over all 230 cases is $+0.000016$ with a case-bootstrap interval of $[-0.000070, +0.000100]$ and $P(\Delta < 0) = 0.355$. The decomposition by bucket shows why the two agree so closely: the 130 rolled-back cases contribute exactly zero by construction, and the two retained buckets very nearly cancel (a mean $+0.000536$ over the 46 shrunk cases against a mean -0.000390 over the 54 low-bucket cases; weighted by bucket size these sum to the $+0.000016$ above). The router therefore returns the source model’s accuracy while deploying roughly a tenth of the edited area, which is a different statement from returning the source model.

Benchmark	Method / bucket	all cases				cases with regions		
		n	HA	BA	BA/HA	n	HA	coverage δ
Cardiac	Router (all)	230	0.013	0.018	1.37	17	0.180	3×10^{-5}
	rollback bucket	130	0	0	—	0	—	0
	shrink bucket	46	0.037	0.091	2.44	14	0.123	7×10^{-5}
	low bucket (depth 3)	54	0.025	0.000	0.00	3	0.446	5×10^{-5}
	retained (both)	100	0.031	0.042	1.37	17	0.180	6×10^{-5}
	fixed-1	230	0.186	0.234	1.26	167	0.256	5.4×10^{-4}
	fixed-2	230	0.164	0.225	1.37	144	0.263	4.6×10^{-4}
	fixed-3	230	0.129	0.159	1.23	101	0.294	3.2×10^{-4}
	fixed-4	230	0.177	0.149	0.84	132	0.308	3.3×10^{-4}
Prostate	Router (all)	93	0.099	0.098	0.99	35	0.264	8.3×10^{-4}
	rollback bucket	42	0	0	—	0	—	0
	mid bucket	24	0.252	0.340	1.35	24	0.252	2.6×10^{-3}
	low bucket	27	0.118	0.036	0.31	11	0.290	5.4×10^{-4}
	retained (both)	51	0.181	0.179	0.99	35	0.264	1.5×10^{-3}
	fixed-1	93	0.242	0.327	1.35	77	0.292	5.4×10^{-3}
	fixed-2	93	0.211	0.283	1.34	69	0.284	5.6×10^{-3}
	fixed-3	93	0.171	0.269	1.57	69	0.231	5.4×10^{-3}
	fixed-4	93	0.262	0.284	1.09	81	0.300	5.6×10^{-3}

Table S15: Three-way accounting of edited area. The *cases with regions* block spreads the same totals over the cases that retain a supra-threshold region, i.e. it rescales the all-case columns by n/n_{regions} ; the within-region BA this implies for the cardiac retained row is 0.248, the value quoted in Sec. S5. Neutral area is the complement of HA+BA and is the largest cell in the all-case columns. BA/HA is comparable across rows; BA−HA is not, because coverage differs by an order of magnitude between the router and the fixed budgets.

S7 CALIBRATION SENSITIVITY PROPAGATED TO RESULTS

Perturbing the calibration split changes the cut-points, the cut-points change the bucket assignment, and the bucket assignment changes what is deployed. This section propagates the perturbation all the way to deployment metrics rather than stopping at the routing labels. For each calibration size we draw 500 subsamples without replacement, refit the cut-points on the subsample, and replay deployment on the full evaluation split.

The quantile convention is not shared across benchmarks and must be copied per domain rather than assumed: the M&Ms router uses `np.percentile(x, [33.33, 66.67])` and the driving router `np.percentile(x, [33, 66])`, both with linear interpolation. On the $n=400$ ACDC fog training scores the two differ in the fourth decimal (0.01249 vs. 0.01257), enough to move cases across a bucket boundary. The medical convention also coincides with `np.quantile(x, [1/3, 2/3])` to five decimals on this split, so the two are not distinguishable from their output here; the convention has to be copied from the source rather than inferred from a value. Refitting the deployed cut-points from the full calibration sets reproduces them exactly on prostate and driving, and on cardiac to within 10^{-8} , a difference that leaves the bucket assignment identical for all 230 cases.

The headline result survives the perturbation and the boundary result does not, which is the same split the other two analyses find. On cardiac every resample at every size tested still beats the retrospective fixed-3, the worst single draw at $n_{\text{cal}}=5$ reaching HA 0.055 against fixed-3’s 0.129, while Dice moves by less than 5×10^{-4} : the router never trades accuracy for the reduction regardless of how the tertiles are fitted. On prostate a fifth to a third of resamples lose to fixed-3, and the mean deployed HA is not monotone in the calibration size (0.121, 0.114, 0.121 at $n_{\text{cal}}=5, 10, 15$), which is what a variance-dominated regime looks like.

On driving we can vary only the rollback quota, for the reason given in Sec. S3; under that restriction the bands narrow monotonically with n_{cal} (fog [0.062, 0.211] at 5 to [0.130, 0.175] at 100; night [0.056, 0.328] to [0.173, 0.228]) but the router’s values sit at or above the band center (snow 0.256 against a mean of 0.250 at $n_{\text{cal}}=100$; these are the fixed-quota values of Sec. S3, not the deployed

Benchmark	n_{cal}	Dice [95%]	HA [95%]	beats fixed-3
Prostate	5	0.7398 [0.7370, 0.7474]	0.121 [0.026, 0.226]	75.6%
	10	0.7393 [0.7370, 0.7421]	0.114 [0.034, 0.196]	66.4%
	15	0.7392 [0.7369, 0.7415]	0.121 [0.069, 0.183]	79.4%
	23 (deployed)	0.7387	0.0994	—
Cardiac	5	0.8490 [0.8488, 0.8492]	0.022 [0.004, 0.048]	100%
	10	0.8490 [0.8488, 0.8492]	0.018 [0.008, 0.045]	100%
	15	0.8489 [0.8489, 0.8490]	0.017 [0.010, 0.033]	100%
	20 (deployed)	0.8489	0.0133	—

Table S16: Deployment metrics under calibration-set subsampling (500 draws per size). The last column is the fraction of resamples whose deployed HA is below that of the retrospective fixed-3 budget. Point estimates are means over the 500 draws; brackets are the 2.5/97.5 percentiles of the draw distribution. Dice is almost invariant to the calibration set on cardiac and moves little on prostate; HA is not.

router’s Table S7 figures). Better calibration does not help when the coordinate itself does not separate at that budget.

S8 SENSITIVITY OF HA TO THE MINIMUM-REGION THRESHOLD

HA counts only disagreement components of at least τ_{area} voxels. The threshold was never swept while the archived run held only the step-4 endpoint mask; the per-step dump of Sec. S9 makes the sweep a pure recomputation over that rerun’s masks. We report $\tau_{\text{area}} \in \{8, 16, 32, 64\}$ around the deployed value of 16.

τ_{area}	cohort HA by step				deployed		
	$k=1$	$k=2$	$k=3$	$k=4$	router	fixed-3	ΔHA
8	0.197	0.176	0.144	0.196	0.024	0.144	−0.120
16	0.186	0.166	0.129	0.179	0.017	0.129	−0.113
32	0.172	0.151	0.111	0.159	0.009	0.111	−0.103
64	0.154	0.138	0.100	0.130	0.005	0.100	−0.095

Table S17: HA on cardiac ($n=230$) recomputed at four minimum-region thresholds. Replayed from per-step checkpoints, so the M&Ms mid bucket is scored at its step-2 state before shrink; the router row here (0.0167 at $\tau_{\text{area}}=16$) is therefore not the deployed value of Table 3 (0.0133). All four rows share this convention, so the comparison across τ_{area} is internally consistent. The fixed- k rows are computed on the per-step masks of the Sec. S9 rerun, which is why the $\tau_{\text{area}}=16$ row differs from Table 1 by 0.002 at $k=2$ and $k=4$, the run-to-run scale that section reports.

HA never increases with τ_{area} in any of the 920 (case, step) rows, but that is an arithmetic property of the definition rather than an observation about edit geometry: the denominator $|D|$ counts all disagreement voxels and does not depend on the threshold, while the numerator sums only components above it, so raising τ_{area} can only remove area from the numerator. Nothing follows from it about whether small components are more often harmful; the region-level contrast of Sec. S1.11 addresses that question directly and on prostate finds the opposite of the “harmful edits are small debris” reading. What does not follow arithmetically, and is the point of the sweep, is that the router’s rank is unchanged across an eightfold change in the threshold, with its margin over the retrospective-best budget staying between −0.095 and −0.120. The count of cases whose edits are scored at all does move: at $k=4$, 148 of 230 at $\tau_{\text{area}}=8$ against 86 at 64 (at $\tau_{\text{area}}=16$, on these same rerun masks, the count is 167/145/102/132 for $k=1..4$, against 167/144/101/132 in Table S15: one boundary case flips at $k=2$ and one at $k=3$, the same cross-execution effect as the HA offset above), so the threshold is in effect a choice of how large an edit has to be before it counts.

S9 VERIFICATION RERUN AND REPRODUCIBILITY OF HA

The archived cardiac run stored only the deployed endpoint mask, so anything requiring the intermediate masks (re-thresholding the minimum region size, a different connectivity, a boundary metric, a qualitative panel at $k < 4$) needed a rerun. We reran EATA fixed-4 on all 230 vendor-B cases, dumping every step, and used the opportunity to log the two probability-derived signals the archived cardiac trajectory lacks (entropy, MSP) plus a consistency signal. The archived run remains authoritative for every main-table deployment result. Analyses that need intermediate masks or per-voxel probabilities are computed on this rerun and are identified where reported.

Layered alignment. A rerun that does not reproduce the archived run is a different experiment, so we check it, but at the layer where a deviation can actually originate. Step 1 is one adaptation step from an identical frozen state, so its gradient norm is fixed by the learning rate, the entropy margin, the Fisher weight, the parameter subset and the preprocessing: we require it to match to 10^{-6} , and it does (2.8×10^{-7} maximum over 230 cases, median 7.5×10^{-9} , i.e. at float32 rounding). Downstream, sliding-window inference and export are not bitwise deterministic across executions, so per-case Dice drifts by up to 1.5×10^{-3} ; we therefore check the quantities the paper actually reports: per-step cohort-mean Dice (maximum deviation 3.6×10^{-5} against a 5×10^{-4} tolerance) and cohort HA. Gating on per-case Dice, which no table prints, would leave only one way to pass: fitting the tolerance to the observed spread.

HA is reproducible to about ± 0.002 , and why. The cohort mean HA at step 4 differs by $+0.0020$ between the two runs, which collapses to -0.0001 once five of 230 cases are excluded. Those five are not noise in the usual sense: HA is a fraction of a region-filtered disagreement set, so when $|D|$ is small a single blob crossing the 16-voxel minimum flips a whole case. The largest single deviation is a case whose step-2 HA is 0.279 in one run and 0 in the other, on a disagreement set of 86 voxels, with region counts 1 and 0. Of the eight cases with $|\Delta \text{HA}| > 0.05$, four cross the size threshold and four keep the same region count while a borderline region changes side. We report ± 0.002 as the run-to-run reproducibility of a cohort HA mean and note that every ΔHA claimed in this paper is at least ten times that scale, with exceptions in Secs. S2.3, S3, S10 and S12, where the differences quoted are two to about four times it. The difference between FragSubset and FragStop on cardiac (0.109 vs. 0.115) is one of them and should not be read as an ordering.

Consistency signals, and one specification that could not be run. A dropout-agreement monitor in the style of the closest related work is not computable on this backbone: the source model is an nnU-Net PlainConvUNet constructed with `dropout_op: None`, so stochastic forward passes return identical outputs and the agreement column would be the constant 1. We therefore report two variants. The primary one perturbs with light augmentations (8 passes, seeds 1000–1007, independent flips of the two spatial axes at $p=0.5$ with the geometry inverted exactly on the logits, brightness in $[0.9, 1.1]$ and Gaussian noise with $\sigma \in [0, 0.05]$ on z-scored intensities); an identity-augmentation self-test asserts agreement exactly 1 before any value is logged. The secondary one injects `nn.Dropout2d` ($p=0.1$) before each segmentation head via forward pre-hooks, which leaves the state dict untouched. The model was never trained under dropout, so that perturbation is out of distribution for it, and the result matches the caveat we recorded in advance: its correlation with HA is 0.047 with an interval containing zero (Table S12), against 0.264 for augmentation agreement. We report it because the caveat was written before the measurement, not after.

Provenance of the cardiac columns in the cross-benchmark table. Entropy, MSP and the consistency signals for cardiac are computed on this verification rerun rather than on the archived run. The signals the two runs share agree to ± 0.002 (n_{reg} 0.596 vs. 0.597, δ 0.516 vs. 0.516, grad-norm 0.148 vs. 0.146), which is what licenses reading the new columns alongside the archived ones.

Provenance of the GraTa-cos row. GraTa’s alignment signal needs two extra backward passes per step, so it was measured in a separate execution of the same $n=124$ trajectories; the stochastic state therefore differs from the run behind the other five rows. That run yields, for the other five signals, 0.594/0.447 (n_{reg}), 0.554/0.510 (δ), 0.119/ -0.298 (entropy), $-0.162/0.283$ (MSP) and 0.261/ -0.035 (grad-norm), a maximum absolute difference of 0.017 from the values printed in

Table 2, with the ranking of the signals and both conclusions the table supports unchanged. We report the values as measured in each run rather than re-running to a single seed.

S10 CONTROLLER, COMPUTE, AND DESIGN PROVENANCE

Algorithm 1 COQR routing for one test case (no labels, and no backward pass for the routing decision itself)

Input: frozen source weights M_0 ; case x ; cut-points (c_{lo}, c_{hi}) ; per-bucket policy Π_d . All fitted once offline on a labeled holdout disjoint from the test stream.

```

1: reset to  $M_0$ ; predict  $\hat{y}_0$ 
2: take one adaptation step; predict  $\hat{y}_1$ 
3:  $u \leftarrow g_d(\phi(\hat{y}_0, \hat{y}_1))$  two masks only; no backward pass
4: if  $u \geq c_{hi}$ : restore  $M_0$ ; return  $\hat{y}_0$  rollback
5:  $b \leftarrow \text{MID}$  if  $u > c_{lo}$  else LOW
6: if  $\Pi_d$  is fixed-depth (medical): run to depth  $\Pi_d[b]$ ; on M&Ms's MID bucket, interpolate half-way back to  $M_0$ ; stop
7: else (driving): continue while  $\Delta n_k \leq 0$ , evaluated only for  $k \geq 2$ , up to the cap  $\Pi_d[b]$  step 1 is always kept

```

Output: the deployed mask, and the number of updates actually deployed.

Table S18: The decision consumes only the two masks $(\hat{y}_0, \hat{y}_1)$ and is taken *after* one adaptation step, so a rolled-back case still pays that step; what rollback saves is the remainder of the budget, while the *Steps* column counts deployed updates, so a rolled-back case contributes 0. Line 6 and line 7 are two different controllers, not two settings of one: the medical routers deploy a constant depth per bucket, and the stop condition present in that code path is vacuous under its indexing (see below).

Component	Per step	Backward passes	Peak extra memory
Fragmentation $\phi(M_0, M_k)$	~ 37 ms	0	negligible
Gradient norm	~ 150 ms	1	2.5 GB
GraTa alignment	—	2	—
Augmentation agreement (8 passes)	$8 \times$ forward	0	negligible

Table S19: Decision-time cost of each signal, over and above the adaptation step itself. The controller adds no backward pass; the gradient-based comparators do. GraTa's wall-clock and peak memory were not measured, and only its backward-pass count is relevant here. As a decision signal it is read-only (Sec. S1.8); where Sec. S12 runs it as an adaptation objective, it is the objective rather than the controller.

How the controller's design was arrived at, and what that costs. Four choices in the deployed controllers were made while an evaluation split was visible, and three constants were not considered at all. We separate them by a criterion a reader can check against the released code: whether the choice has a command-line option and an archived alternative that was run and scored, or is a hard-coded module constant with neither.

Selected on M&Ms with evaluation-split visibility. Three choices were settled in an exploratory sweep in which twelve router variants were evaluated on the vendor-B test split, the same $n=230$ split the main table reports. That sweep moved the scoring coordinate from the two-term risk of Eq. 2 to the disagreement ratio δ . At the cut-points fitted for it, the two-term risk did not partition this split at all: all 230 cases fell in a single bucket, so those variants reduce to fixed-depth policies, and the first of them reproduces the fixed-1 row of Table 1 to within run-to-run noise (HA 0.186 at depth 1; the two differ by 10^{-4} , well inside the ± 0.002 of Sec. S9). The budget-axis controllers read r_k against a fixed sign threshold rather than tertiles and are unaffected. With δ the same architecture gave a three-way split at HA 0.066, and later variants reached as low as the 0.040 of the final architecture below. The sweep also settled whether the middle bucket shrinks, halts or is delayed; and it settled the depth of the low bucket, through an option named for a stopping policy (default / delay4 / nostop) whose three settings, under the fixed-depth behavior documented above, amount to deploying the low bucket at depth 1, 3 or 4. The archived variants isolating that option reach HA 0.044 (delay4,

the setting carried into the deployed design) and 0.060 (nostop). Both isolate the option under the online sliding-window quantile mode (quantile_warmup=15, cut-points computed on the test stream) rather than the seed-only calibration used everywhere in this paper (Sec. S1.7); the same delay4 setting under seed-only calibration is the 0.040 of the final architecture. We name the option as it appears in the code rather than as it reads, because a reader who saw only the flag would take it for a data-driven stopping rule and not for a choice of integer depth. The signal-validity argument of the main text does not cover these choices and should not be read as covering them: that argument establishes where a coordinate ceases to be usable, and on cardiac neither coordinate is near its boundary, and n_{reg} is in fact the stronger of the two on that benchmark (Table 2). The reason δ is deployed there is the sweep, not the boundary.

Compared on ACDC with evaluation-split visibility. One further choice was made while looking at the driving evaluation split, and it is smaller in both kind and degree: a single two-way comparison, not a search. The tertile upper cut-point was compared at $P66$ and $P75$ on val, and $P66$ is deployed; it is better on all four conditions (mean HA 0.203 against 0.225) at mIoU differences of 0.0012 or less. $P66$ is the defining value of a tertile, and the medical routers use the same convention at $P66.67$, so the comparison did not move the deployed configuration off the default. An author who never looked at val would have deployed the same cut-point. And what $P66$ buys is a larger hard bucket (18/21/29/43 against 9/17/25/32), i.e. more rollback rather than better separation, which is what Sec. S3 independently finds at this budget and another instance of the signal ordering where to do less. Neither qualification makes the comparison label-blind, and we record it as an exposure.

A vacuous branch in the medical router. The medical router’s code path retains a stop condition inside its per-step loop, but the condition cannot fire selectively: the region-count history is appended at the end of the step body and read at the start, so at step k it holds $k - 1$ entries and the test $\text{CHOOSE}(\cdot) < k$ is satisfied for every case regardless of the geometry. The consequence is the fixed depth per bucket documented in Table S1 and measured in Table S2: the fragmentation state enters the bucket assignment and not the depth. Every deployed number in this paper is a measurement of that behavior and none of them changes. The standalone FragStop controller reported in the ablations evaluates the same condition on a complete archived trajectory and is unaffected; and the matched-control experiment (Sec. S3) varies only the score that assigns cases to buckets, so what it measures is the whole of the signal’s contribution rather than part of it. Correcting the indexing yields a different controller, whose per-bucket depths, tertiles and middle action would all have to be re-selected; we have not run it and do not report a corrected version.

Unconsidered constants. The shrink weight $\alpha=0.5$ is the default value of the configuration object and was never varied. The ACDC step caps are likewise hard-coded module constants with no command-line option and no archived alternative: MID_MAX_STEPS=4 carries over from the medical configuration and LOW_MAX_STEPS=8 is set equal to that domain’s own budget. These are unconsidered constants rather than tuned ones.

The thresholds were subsequently refit on a disjoint vendor-B calibration split of twenty cases and are clean (the improvement from 0.040 to 0.013 between the final architecture and the reported result is entirely that recalibration: the two runs agree on every recorded configuration field and differ only in the cut-points), but recalibrating thresholds does not undo a design selected with the test split visible.

We report this rather than presenting the architecture as given. Three things bound how much it can be worth. The transfer results fix the template and refit thresholds, with the coordinate and per-bucket actions set per domain, but they are not uniformly clean and we now separate them. Prostate OOD-all and Prostate158 took no part in any of the selections above: no coordinate, action, depth or cut-point on either was chosen while their evaluation splits were visible, and Prostate158 in addition freezes the entire M&Ms design rather than refitting a coordinate. Those two are unexposed. ACDC is not: the cut-point comparison above was made on its evaluation split, so the driving numbers carry an exposure of their own, smaller than the M&Ms sweep but not absent. What the driving section actually rests on is unaffected. The monotone $\text{TENT} \rightarrow \tau\text{-stop} \rightarrow \text{router}$ cascade (Sec. 5.3) and the signal-validity boundary at the aggressive budget (Sec. 5.3) are descriptive observations about archived trajectories, and neither depends on how the cut-point was picked. The bound this paragraph offers, however, now covers two benchmarks rather than three. The matched-control experiment of Sec. S3 compares scoring coordinates at a fixed architecture, so the margin it measures is not attributable to the architecture search. And the direction of the exposure is visible in Sec. S7: the

cardiac advantage survives calibration sets of five cases, which is not what a design fitted to 230 evaluation cases would be expected to do. A fourth bound is available offline, and we ran it after the fact. The sweep compared the two-term risk against δ ; it never considered n_{reg} , which is therefore a coordinate the selection could not have favored. Substituting it into the deployed architecture, with the same actions, bucket sizes and calibration split and only the ranking score changed, gives HA 0.0366 at Dice 0.8488, against δ 's 0.0133 at 0.8489. The un-swept coordinate is worse than the swept one, as one would expect, but it still separates from every matched control in Sec. S3 (augmentation agreement 0.0451, entropy 0.0537, MSP 0.0567, random 0.0605): the margin over the alternatives is not an artifact of having searched, because a coordinate that was never searched retains most of it.

That experiment also supplies a reason for the cardiac choice that does not require the sweep. Fitted natively rather than at matched quota, n_{reg} has no resolution on this benchmark: its values on the twenty calibration cases are 0 ($\times 10$), 1 ($\times 6$) and 2 ($\times 4$), so its tertiles collapse to 0 and 1 and the middle bucket is empty. The resulting router rolls back 167 of 230 cases and reaches HA 0.0039 with BA 0.0000. It wins on HA by abstaining almost everywhere, which is the degenerate solution Sec. 5.4 rules out. δ takes twenty distinct values on the same twenty cases and yields three non-empty buckets. **That contrast is visible in the calibration set alone, with no evaluation data**, so an *a priori* rule would have selected δ on cardiac without the sweep: do not route on a coordinate whose tertiles degenerate on the calibration split. We did not apply such a rule at the time, and we are not claiming retrospectively that we did; we report it because a reader is entitled to know whether the choice is defensible on evidence that was available without looking at the test split, and here it is.

A frozen-design test on a benchmark outside the sweep. The bounds above all ask whether the search could have produced the reported margin. A different question is whether the design that came out of it transfers at all, and that one admits a direct test. We froze the M&Ms router entirely, including the scoring coordinate δ , the tertile structure and the per-bucket depths 0 / 2 with shrink / 3, and ran it once on Prostate158 ($n=126$), a benchmark that took no part in the sweep and that shares no cases with any split reported elsewhere in this paper. Only the cut-points were refit, on a 32-case holdout of that benchmark, by the same seed-only procedure. Nothing was tuned, and the run was scored once.

Method	Dice	HA	BA	Steps
M_0 (no adapt, ref.)	0.7985	0	0	0
fixed-1	0.8023	0.311	0.439	1.00
fixed-2	0.8031	0.249	0.434	2.00
fixed-3 (<i>deployable = retrospective</i>)	0.8023	0.228	0.376	3.00
fixed-4 (<i>default</i>)	0.8004	0.298	0.382	4.00
Frozen M&Ms design	0.8000	0.139	0.224	1.76

Table S20: The frozen-design test on Prostate158 ($n=126$ evaluation cases). The design is frozen from M&Ms; only the cut-points are refit, on a 32-case holdout of this benchmark. The router rolls back 39 of 126 cases. The deployable and retrospective budgets coincide here at $k=3$.

Three readings, one of which is unfavorable. First, the reduction clears the strongest reference available: ΔHA against the retrospective-best fixed-3 is -0.089 $[-0.124, -0.055]$ with $P=1.000$, at ΔDice -0.0023 $[-0.0050, +0.0001]$, an interval that includes zero, and the router sits above M_0 ($+0.0015$ $[+0.0007, +0.0023]$). The Dice interval only barely includes zero ($P(\Delta < 0)=0.972$), so we read it as an interval containing zero rather than as an absence of cost. Second, the three reference tiers collapse to two here: the 32-case holdout selects $k=3$, which is also the evaluation optimum, so Tier-A and Tier-B coincide. This is the situation ACDC already poses (Sec. 5.3), a budget that is both optimal and selectable, and the answer is the same, with a larger margin (-0.089 against ACDC's -0.048). Third, the composition of the deployed edits does not improve: BA/HA is 1.61 for the router against 1.28–1.74 across the fixed- k ladder, so the HA reduction comes from deploying less rather than from deploying better. That is a further independent replication of the limitation stated in Sec. 5.2 and quantified in Sec. S6.

As elsewhere we report retained-only figures so the abstention component is visible: over the 87 retained cases HA is 0.201 at coverage $\delta=1.02 \times 10^{-3}$, against 0.139 and 7.02×10^{-4} over all 126. Unlike cardiac, where only 17 of 230 cases retain a supra-threshold region, here the set of cases with

scorable regions coincides exactly with the retained set: every case this router does not roll back deploys an edit large enough to be scored.

What the frozen test does and does not bound. It establishes that the design transfers: applied unchanged to a benchmark it could not have been fitted to, it clears that benchmark’s own retrospective optimum. It does not establish that the M&Ms number is unbiased, and the three-benchmark comparison says as much. The relative HA reduction is 90% on M&Ms, 42% on prostate OOD-all and 39% on Prostate158. The two benchmarks outside the sweep agree with each other and not with the one inside it. Three accounts are consistent with that pattern and we cannot separate them here. The rollback rates differ in the same direction (57%, 45%, 31%), so part of the gap is that M&Ms declines more; but the relation is not proportional. Twelve points more rollback than prostate buys forty-eight points more reduction, so rollback rate alone does not account for it. The edit sets also differ by an order of magnitude in coverage (3×10^{-5} on cardiac against 7×10^{-4} here), and HA is a fraction of that set, so a sparse edit set is easier to drive near zero. Selection effect is the third account. On Prostate158 a matched-bucket control now bears on it directly: at the same quota and the same depths the ranking is not better than random (Sec. S3), so the reduction there is carried by the quota rather than by case selection. Whether the same is true of the 90% on M&Ms is not settled here, and we do not claim to have excluded a selection effect on the benchmark the design was chosen on. A reader who treats 39%–42% as the unbiased estimate of what this design and its template buy, and 90% as an upper bound specific to the benchmark it was selected on, is reading the evidence the way we do.

A design search on a held-out split, which we did not run, would settle the question; we list it as the first thing to redo rather than as a limitation that argues itself away.

The verification rerun of Sec. S9 took ~ 27 s per case for four steps with both consistency signals; materializing the shrink action for all 230 cardiac cases took 2.3 s per case.

S11 A TRANSFORMER BACKBONE: WHAT PORTS AND WHAT DOES NOT

A source model below its own usability line. A 3D Swin-UNETR reached 0.6861 source-domain Dice against nnU-Net 2D’s 0.8916, triggering a pre-registered fallback to a 2D transformer. Three 2D runs placed their final checkpoints at 0.8394, 0.8130 and 0.8220. The best checkpoint cleared the 0.8416 fallback line every time and the final checkpoint never did. Under the pre-registered convention of reporting the final checkpoint, this source model therefore sits 0.020 below it. On vendor-B its M_0 Dice is 0.630 against 0.849, and it segments non-cardiac structures that nnU-Net does not (at lower right in Fig. S6, where the ground truth labels nothing). Its step-1 edit volume on vendor-B is $\delta = 1.45 \times 10^{-4}$, 0.27 of nnU-Net’s 5.45×10^{-4} , at the learning rate the pre-registration fixed in advance. What follows is conditional on a source model weaker than the main text’s.

One coordinate transfers, the other does not. At the pre-registered $\tau_{\text{area}} = 16$, n_{reg} predicts HA at $\rho = 0.669$ [0.446, 0.836], clearing the 0.4 bar; δ reaches 0.151 [0.081, 0.206]. The criterion requires both, so it is not met, and it is δ that fails. The interval is wide because HA is non-zero in only 9 of 920 case-steps (8 of 230 cases), leaving the case-clustered bootstrap little to resample. We read only that its lower bound clears 0.4, and claim no ordering against the nnU-Net line, whose narrower interval [0.532, 0.657] lies wholly inside this one. One reservation covers this whole section. Cohort HA never exceeds 0.0053 here at any k , so almost none of the edited area is scored as harmful. Every verdict in this section is therefore read off a set of harmful edits that barely exists.

The edits have no coherent core. Matching on M_0 within [0.60, 0.80] (each backbone’s own cases in the band; the two distributions overlap on only 7), the largest disagreement component averages 345 voxels on nnU-Net (43 cases) against 3.5 on the 2D transformer (115 cases), a factor of 97.7. The share of changed voxels in singletons is 0.33 against 0.83. Lowering the threshold recovers nothing: ρ falls monotonically to 0.574, 0.560 and 0.337 at $\tau_{\text{area}} = 8, 4, 2$. Here 16 sits close to the boundary between edit and noise. This is independent support for the main text’s choice, from a backbone whose edit geometry is entirely different.

Consequences. δ is a share of changed voxels and under scatter mostly measures noise, whereas n_{reg} counts only supra-threshold components: the same τ_{area} filters for one and does nothing for the

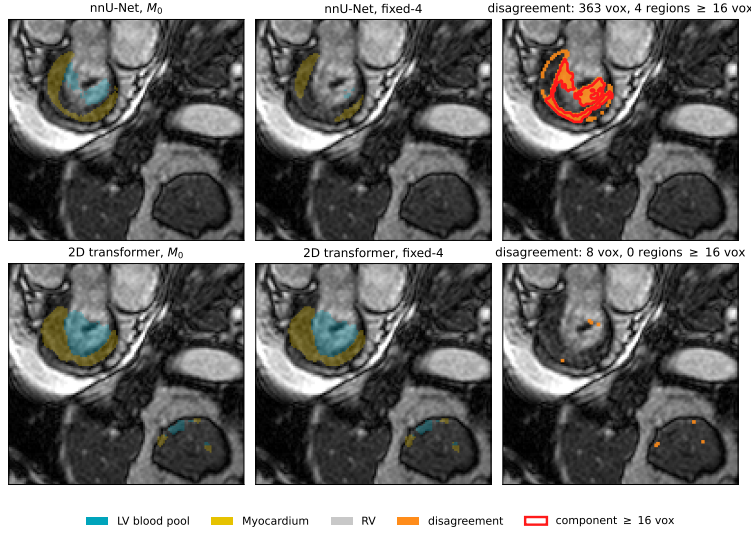


Figure S6: Edit geometry across backbones, one case (G1N6S7_frame12, slice 8; M_0 Dice 0.75 nnU-Net / 0.63 2D transformer). Columns: M_0 , fixed-4, and their disagreement (orange), with 3D components of at least 16 voxels outlined in red. Over the volume the nnU-Net edit forms 5 such components and the transformer edit forms none (largest 2 voxels); on the slice shown, 4 and none. On this case fixed-4 lowers nnU-Net Dice by 0.016 at HA 0.77, which is what the scorable components capture. The two edits differ in volume (363 against 8 voxels here) and in structure; Sec. S11 rests on the second. Orange is dilated by one pixel for visibility, so it overstates the true area; the counts in the titles are undilated. The grey legend entry has no instance here: neither model predicts RV on this slice, and the ground truth has none. Case and slice selected by the rule of Sec. S1.15: the median transformer-side disagreement volume among the 7 cases with both M_0 in $[0.60, 0.80]$, and the slice carrying the most nnU-Net supra-threshold disagreement area. The case is conservative for the claim (nnU-Net largest component 116, below the band mean of 345).

other. The deployed M&Ms router scores on δ and the prostate router on n_{reg} , so on this backbone it is the cardiac coordinate that fails to port. The router’s $+0.00044$ $[+0.00031, +0.00057]$ Dice over fixed-4 is no gain: router- M_0 is -0.00004 $[-0.00011, +0.00003]$ while fixed-4- M_0 is -0.00048 $[-0.00061, -0.00036]$. It avoids a loss rather than earning a gain: the router sits at M_0 here as on cardiac (Sec. 5.1), and the difference is that fixed-4 falls below it.

S12 TWO OTHER ADAPTATION OBJECTIVES: WHAT THE CONCLUSIONS SURVIVE

What this section is for. The main text claims that *whether* a step should be deployed is independent of *how* the trajectory is optimized. This section tests that on two published objectives: GraTa (Chen et al., 2025), a gradient-alignment method for medical segmentation, and DeYO (Lee et al., 2024a), a classification method whose sample selection we translate to slices. Both generate fixed- k trajectories on the benchmarks, splits and protocol of the main tables, and the router of Sec. S3 is replayed offline on each. Criteria V0–V4 were fixed before any run and are reported whether or not they hold (Table S21); one arm and one reduction added after the data existed are marked as such.

What was ported, and what deviates. GraTa follows the official release step for step: an entropy gradient at θ and a unit-step perturbation $\theta' = \theta - \nabla_{\text{ent}}$ with no learning rate; a target averaged over six weak views (identity, two flips, three rotations) after inverting each view’s geometry on the probabilities; a single strongly augmented view; and an Adam step along the gradient ∇_{con} of the consistency cross-entropy $L_{\text{con}}(\theta')$ with $\eta = 10^{-4} \cdot \frac{1}{4}(\cos + 1)^2$, where $\cos$ is the cosine between the two gradients. InstanceNorm affine parameters, the shared optimizer and budget, and the episodic reset follow Sec. S1.3 rather than each method’s official choice (the official DeYO, for instance, uses

SGD), so that the only intended difference between objectives is the objective. Five deviations are specific to this section. (i) Head: the official head is sigmoid and two-class; ours is softmax over C classes ($C=4$ cardiac, $C=2$ prostate), and the mean of six softmax maps is *not* renormalized. (ii) Steps: the official method takes one step per case; we run $K=4$ to obtain a ladder, so fixed-1 is native GraTa. (iii) Gamma: the five augmentation probabilities, $0.75/0.75/0.75/0.5/0.5$, are the official ones, but our gamma transform keeps `retain_stats` true where the official release leaves the library default of false. (iv) Blur: the official blur is applied per channel with probability 0.5 inside a sample selected with probability 0.5; ours omits the per-channel draw, so on single-channel MRI a strong view is blurred with probability 0.5 rather than 0.25. This was an oversight in the port, found only when checking against the official source, and its effect is measured below. (v) Intensity range: the strong augmentation acts on z -scored intensities rather than the official $[0, 1]$ fundus range, so multiplicative brightness is a gain change and a noise variance of 0.05 is $\sigma \leq 0.22$ in z units.

The official-lr arm does not move. On cardiac vendor-B, GraTa at fixed-1 scores 0.8489 Dice against M_0 's 0.8489 and TENT's 0.8507; on prostate, 0.7364 against 0.7379 and 0.7518. The pre-registered V0 requires GraTa to reach at least TENT and not to fall more than 0.01 below M_0 . The second half holds on both benchmarks (+0.0000 and -0.0015); the first fails on both. The published GraTa gains 5.9 Dice points over its source model across five fundus domains and beats TENT and SAR there, so the direction does not reproduce: what appears is not a loss but no effect. Step-1 δ is 2.26×10^{-5} on cardiac and 3.27×10^{-4} on prostate, 0.041 and 0.062 of what the main text's objective produces on the same data, and the effective learning rate, 1.3×10^{-5} and 2.1×10^{-5} , is a thirty-eighth to a twenty-fourth of the shared base.

Why the learning rate collapses. The multiplier $\frac{1}{4}(\cos+1)^2$ is small when the two gradients disagree, and here they disagree systematically: $\cos$ averages -0.300 on cardiac ($n=920$ case-steps, s.d. 0.206) and -0.108 on prostate ($n=372$, s.d. 0.199). Of five non-degeneracy checks, four pass; the fifth, a multiplier spanning at least $[0.2, 0.9]$, fails, since it never exceeds 0.53. Part of the mechanism is the source model's confidence. At step 1 on the evaluation cases ($n=230$ cardiac, 93 prostate), the six-view target at θ' has mean per-pixel entropy 0.0039 and 0.0150, 4.9 and 5.2 times that of the identity view (0.0008, 0.0029). Averaging near-one-hot maps that disagree only at boundaries gives a softer target, so descending the consistency cross-entropy pushes *against* entropy descent. That explains about half of the sign: hardening the target to one-hot moves step-1 $\cos$ from -0.295 to -0.170 on cardiac and from -0.100 to -0.051 on prostate, still negative, while the entropy gradients of the plain and strongly augmented inputs align at $+0.53$ and $+0.43$ on average. The remainder is not identified: target hardening and the official blur of (iv) were tested and neither closes it, and (v) was measured but not ablated. Deviation (v) contributes: the backbone was trained under this augmentation family, and a strong view leaves 0.963 and 0.908 of labels unchanged on the union of foreground, supplying little of the disagreement GraTa expects to exploit. Deviation (iv) runs the other way, and only slightly: with the official blur $\cos$ becomes marginally more negative (-0.300 , -0.109), so the oversight made GraTa's steps slightly larger, not smaller. This is a statement about GraTa's applicability to a very confident backbone, not about its correctness.

Given a comparable learning rate, the objective does produce edits, and on prostate it does harm. To separate "the trajectory did not move" from "the objective does nothing on this backbone", a supplementary arm, *GraTa (shared lr)*, uses base learning rate 5×10^{-4} , the value DeYO, TENT and EATA already use, and keeps the cosine modulation as part of the objective. It was added **after** the official-lr arm had run and is not pre-registered, since the pre-registration says to keep GraTa's own rule; it is therefore reported beside the official-lr arm and never in place of it. Cardiac HA rises from 0.008/0.019/0.026/0.041 to 0.061/0.102/0.119/0.127, a factor of 3.1 at $k=4$, and δ rises to 0.197 and 0.311 of the reference, still well short of it, since $\cos$ is unchanged by construction. Dice does not follow. On cardiac it stays at M_0 (0.8491 at $k=1$, 0.8493 at $k=4$); on prostate it falls monotonically, $0.7296 \rightarrow 0.7225 \rightarrow 0.7165 \rightarrow 0.7099$, from an M_0 of 0.7379. V0's first half still fails in both arms.

A slice-level adaptation of DeYO, and one reduction changed after the fact. The official DeYO has no segmentation experiment, so this is a translation and is named as one throughout, as Sec. S1.3 does for EATA. The sample is a slice, and the per-pixel drop $p(x)_{\hat{y}} - p(x')_{\hat{y}}$ under a block shuffle x' is taken against the original pseudo-label. Thresholds are the official defaults converted to C classes:

Table S21: The four pre-registered criteria on two objectives, two benchmarks, and the supplementary shared-learning-rate arm; criteria as defined in the text. $\rho(\delta)$ and $\rho(n_{\text{reg}})$ are the pooled case-clustered Spearman correlations against HA on which V1 is decided. V4 is the share of the router’s HA reduction reproduced by a matched-quota control and is reported, not tested. GraTa (shared lr) and the foreground reduction of the DeYO drop were both introduced after the data existed.

Objective	Benchmark	$\rho(\delta)$	$\rho(n_{\text{reg}})$	V1	V2	V3	V4 random
GraTa (official lr)	cardiac	0.362	0.737	×	×	✓	0.742
GraTa (official lr)	prostate	0.734	0.883	×	×	✓	0.855
GraTa (shared lr)	cardiac	0.564	0.719	×	×	✓	0.720
GraTa (shared lr)	prostate	0.802	0.720	×	×	✓	0.752
DeYO (slice-level)	cardiac	0.475	0.463	✓	×	✓	0.694
DeYO (slice-level)	prostate	0.372	0.367	×	✓	×	0.821

$\tau_{\text{Ent}}=0.5 \ln C$, $\text{Ent}_0=0.4 \ln C$, $\tau_{\text{PLPD}}=0.2$. Reducing the drop over *all* pixels, as pre-registered, selected *no slice at all* on either benchmark at any of the four steps (2434 cardiac and 1050 prostate evaluation slices per step), leaving both DeYO trajectories identical to M_0 . The cause is class imbalance, not the shuffle: 96.9% and 96.7% of a slice’s pixels are pseudo-labelled background, whose near-zero drop dilutes the slice mean to at most 0.124 and 0.143, while on pseudo-label foreground the median drop is 0.989 and 0.981. We therefore reduce over foreground pixels. **This is a post-hoc change to a pre-registered quantity**, authorized and recorded as such. Selection stays at slice level, not the pixel-level variant the pre-registration excludes, and slices with no pseudo-label foreground get a drop of zero. After the change, 74.9% and 99.0% of slices are selected; exactly one cardiac case-step out of 920 still selects nothing (mean drop 0.070), and none on prostate.

Budget non-monotonicity does not reproduce. V2 asks that cohort HA be non-monotone in k with HA(4) above the ladder minimum. It holds in one of six cells. In the other five HA rises monotonically with the minimum at $k=1$, including both GraTa arms, and since the shared-lr arm tripled HA and stayed monotone, this is not an artifact of small steps. The passing cell is the DeYO translation on prostate, 0.2405/0.2954/0.3090/0.3038, where a 0.0052 fall from $k=3$ to $k=4$ meets both conditions. The non-monotone budget of the main text depends on the objective and the horizon it uses; under these two objectives the cheapest budget is also the safest.

The signal reproduces in one cell of six. V1 asks that the pooled case-clustered ρ of both δ and n_{reg} against HA fall in $[0.4, 0.6]$, a two-sided band, unlike the one-sided 0.4 bar of Sec. S11; each was fixed before its own runs and neither is revised here. Only the DeYO translation on cardiac satisfies both (0.475 $[0.375, 0.561]$ and 0.463 $[0.369, 0.549]$). The other five leave the band in both directions (Table S21). Above it lie six of the eight GraTa values: n_{reg} in all four cells and δ on prostate in both arms. Below it lie δ in the official-lr arm on cardiac (0.362) and both values for the DeYO translation on prostate. So three of the five failures are failures of a correlation stronger than pre-registered, one (the DeYO translation on prostate) is one where it has weakened, and one is mixed: in the official-lr arm on cardiac δ falls below the band while n_{reg} sits above it. Raising the step size moves cardiac $\rho(\delta, \text{HA})$ from 0.362 into the band at 0.564, which suggests the low official-lr value reflects a trajectory that barely moves rather than a signal that does not exist. Under the pre-registered criterion, the claim that the fragmentation signal reproduces across adaptation objectives is not established.

The router wraps these trajectories; its ranking is not what earns the reduction. With cut-points refit on each arm’s own calibration split and the actions of Sec. S3 unchanged (cardiac 3/2/0 with the mid bucket at depth 2 and no shrink, prostate 1/1/rollback), the router lowers HA in all six cells. V3’s Dice condition, that the interval on the cost of routing end at most $+0.005$, holds everywhere but the DeYO translation on prostate ($+0.0178 [-0.0007, +0.0389]$, the cell where fixed- k gained the most Dice and the router declines it), so the related-work claim that a fragmentation router wraps a GraTa-optimized trajectory as it wraps ours is supported in the narrow sense that wrapping is possible and nearly free. On the shared-lr arm on prostate the router is 0.0284 $[0.0140, 0.0443]$ Dice *better* than fixed-4, because fixed-4 there is harmful. A matched-quota random control (bucket sizes and actions kept, ranking replaced by 1000 permutation draws) removes most of the weight from that

support: it reproduces 69.4% to 85.5% of the router’s HA reduction across the six cells, and ranking by n_{reg} reproduces 79.2% to 100.0%. On prostate the two non-rollback buckets share an action, so the router reduces to declining a fixed quota of cases, which a random quota of that size does comparably well. The defensible reading is that these trajectories *can* be wrapped without losing Dice, not that the benefit of wrapping them comes from ranking cases by fragmentation.

Two disclosures. First, an earlier round of these runs was discarded. The strong augmentation seeded only NumPy’s global generator while two of the five transforms draw from Python’s `random`, and the per-step seed omitted the case identifier, so every case got the same augmentation draw and, in the DeYO translation, the same block permutation. No conclusion changed when this was fixed, but that round was not reproducible from its own recorded seeds and is kept only as a record. In it, a three-case pilot reported V0 as satisfied on cardiac (0.8624 against 0.8529) where all 230 cases give 0.8489 against 0.8507. A pilot can refute a threshold but cannot establish one. Second, Sec. [S11](#) faced a comparable question and answered it differently. Its pre-registration committed to leaving the learning rate alone even at a large δ mismatch, and its measured ratio (0.27) never approached the one here (0.041), so it has no shared-lr arm. The sources also differ: there δ follows from the backbone’s edit geometry, so changing the learning rate would change what is being measured; here it follows from GraTa’s own learning-rate rule, so the shared-lr arm ablates that rule’s base factor and nothing else. `cos` is unmoved by the base rate (−0.306 against −0.300 on cardiac, −0.115 against −0.108 on prostate), so the arm should not be read as a rate tuned when convenient.